

SIM1001 Basic Mathematics

SIX1013 Fundamental of Advanced Mathematics

Part II LINEAR ALGEBRA

Chapter 1 Linear Systems and Gaussian Elimination

1.1 Linear Systems and their solution

Definition 1.1.

A **linear equation** in n variables $x_1, x_2, \dots, x_n$ has the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

where $a_1, a_2, a_3, \dots, a_n$ and b are real constants. An $a_i, 1 \leq i \leq n$, is called a **coefficient**, and the n variables in the linear equation are also called the **unknowns**.

Example 1.1.

1. The equations $4x - 2y = 1$, $x - 4y + 7z = 5$, $x_1 + x_2 - 2x_3 + x_4 + 0.5x_5 = 0$ and $x_1 + x_2 + \dots + x_n = 1$ are linear equations.
2. The equations $xy = 1$, $3x^2 + 4y^2 = 5$, $\sin x + \cos y = 1$, and $x - e^y = 0$ are non-linear.

Definition 1.2. Given n real numbers $s_1, s_2, s_3, \dots, s_n$. we say that $x_1 = s_1, x_2 = s_2, \dots, x_n = s_n$ is a **solution** of a linear equation $a_1x_1 + a_2x_2 + \dots + a_nx_n = b$ if the equation is satisfied when we substitute the values into the equation accordingly. The set of all solutions of the equations is called the **solution set** of the equation and an expression that gives us all these solutions is called the **general solution** of the equation.

Example 1.2.

1. Consider the linear equation $4x - 2y = 1$.
 $x = 0, y = -1/2$ is a solution of $4x - 2y = 1$. $x = 1/4, y = 0$; and $x = 1, y = 3/2$ are also solutions of $4x - 2y = 1$. The linear equation $4x - 2y = 1$ has infinitely many other solutions.

$x = t, y = 2t - \frac{1}{2}, t \in \mathbb{R}$, (t is an arbitrary parameter) is called the general solution of the linear equation.

The solution set of the linear equation is $\left\{ (x, y) \mid x = t, y = 2t - \frac{1}{2}, t \in \mathbb{R} \right\}$ or

$$\left\{ \begin{pmatrix} t \\ 2t - \frac{1}{2} \end{pmatrix}, t \in \mathbb{R} \right\}.$$

We also can write the general solution as $x = \frac{s}{2} + \frac{1}{4}, y = s, s \in \mathbb{R}$ (s is an arbitrary parameter).

2. The general solution for the linear equation $x - 4y + 7z = 5$ is by $x = 5 + 4s - 7t, y = s, z = t, s, t \in \mathbb{R}$ (s and t are arbitrary parameters).

Definition 1.3. A finite set of m linear equations in n unknowns $x_1, x_2, \dots, x_n$ is called a **system of linear equations** (or **linear system**):

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_n \end{aligned}$$

where a_{ij} and b_i , $1 \leq i \leq m$, $1 \leq j \leq n$ are real constants. An a_{ij} is called a **coefficient**, b_i is called a **constant** and x_j is called an **unknown** (or **variable**).

Given n real numbers $s_1, s_2, s_3, \dots, s_n$. $x_1 = s_1, x_2 = s_2, \dots, x_n = s_n$ is a **solution of the system** if it is a solution of **every** equation in the system. The set of all solutions of the system is called the **solution set** of the system and an expression that gives us all these solutions is called the **general solution** of the system.

Example 1.3. Consider the system of linear equations: $x - 4y + 7z = 5$
 $2x + 15z = 10.$

$x = 5, y = 0, z = 0$ is a solution of the system because it satisfies both equations, but $x = 9, y = 1, z = 0$ is not a solution of the system because it does not satisfy $2x + 15z = 10$.

Remark 1.1:

NOT ALL **system of linear equations** have solutions. For example: the system of linear equations

$$\begin{aligned} x - 2y + 3z &= 1 \\ -2x + 4y - 6z &= 5 \end{aligned}$$

has no solution.

Definition 1.4. A system of linear equations that has at least one solution is called **consistent**. A system that has no solution is said to be **inconsistent**.

Remark 1.2:

Every system of linear equations has either no solution, exactly one solution or infinitely many solutions.

Definition 1.5.

A system of m linear equations in n unknowns $x_1, x_2, \dots, x_n$:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_n \end{aligned}$$

is equivalent to the matrix equation $AX = B$

where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$ is the coefficient matrix, and

$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ and $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$ are the column vectors of the unknowns and of the constants, respectively.

If $B = 0$ (that is, $b_i = 0$ for all i), the system is said to be **homogeneous**.

If $B \neq 0$ (that is, there exists a $b_i \neq 0$), the system is said to be **non-homogeneous**.

This system of m linear equations can be represented by a rectangular array numbers

$$[A | B] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} & b_m \end{array} \right]$$

This array is called the **augmented matrix** of the system.

Example 1.4: The array $\left[\begin{array}{ccc|c} -1 & 1 & 2 & 2 \\ 3 & -1 & 1 & 6 \\ -1 & 3 & 4 & 4 \end{array} \right]$ is the augmented matrix of the system of linear equations

$$\begin{aligned} -x_1 + x_2 + 2x_3 &= 2 \\ 3x_1 - x_2 + x_3 &= 6 \\ -x_1 + 3x_2 + 4x_3 &= 4. \end{aligned}$$

1.2 Elementary row operations and Row Echelon Forms

Definition 1.5: Elementary row operations

A matrix A is said to be **row equivalent** to a matrix B if B can be obtained from A by a finite sequence of the following operations called **elementary row operations**:

[E1]: Interchange the i th row and the j th row: $R_i \leftrightarrow R_j$.

[E2]: Multiplication of a row by a nonzero constant λ : $R_i \rightarrow \lambda R_i$, $\lambda \in \mathbb{R} \setminus \{0\}$.

[E3]: Addition of a constant multiple of one row to another: $R_i \rightarrow R_i + \beta R_j$, $\beta \in \mathbb{R}$.

Example 1.5:

$$[E1]: \left[\begin{array}{cccc} -1 & 1 & 2 & 2 \\ 3 & -1 & 1 & 6 \\ -1 & 3 & 4 & 4 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{cccc} -1 & 3 & 4 & 4 \\ 3 & -1 & 1 & 6 \\ -1 & 1 & 2 & 2 \end{array} \right].$$

$$[E2]: \left[\begin{array}{cccc} -1 & 1 & 2 & 2 \\ 3 & -1 & 1 & 6 \\ -1 & 3 & 4 & 4 \end{array} \right] \xrightarrow{R_3 \rightarrow -3R_3} \left[\begin{array}{cccc} -1 & 1 & 2 & 2 \\ 3 & -1 & 1 & 6 \\ 3 & -9 & -12 & -12 \end{array} \right].$$

$$[E3]: \left[\begin{array}{cccc} -1 & 1 & 2 & 2 \\ 3 & -1 & 1 & 6 \\ -1 & 3 & 4 & 4 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 + 3R_3} \left[\begin{array}{cccc} -1 & 1 & 2 & 2 \\ 0 & 8 & 13 & 18 \\ -1 & 3 & 4 & 4 \end{array} \right].$$

Theorem 1.1: If augmented matrices of two systems of linear equations are row equivalent, then the two systems have the same set of solutions.

Definition 1.6: Row-Echelon Form

A matrix is said to be **row-echelon form** if it has the following properties:

1. If there are any zero rows (a zero row is a row with all entries are zeros), then they are grouped together at the bottom of the matrix.
2. In any two successive non-zero rows (a non-zero row is a row consists at least one non-zero entry), the first non-zero number in the lower row occurs farther to the right than the first non-zero number in the upper row. (The first non-zero number in a row is called the **leading entry** of the row.)

$$\begin{pmatrix} 0 & \cdots & 0 & a_{1j_1} & * & \cdots & * & * & * & \cdots & * & * & * & \cdots & * & * & * & \cdots & * \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & a_{2j_2} & * & \cdots & * & * & * & \cdots & * & * & * & \cdots & * \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & a_{3j_3} & * & \cdots & * & * & * & \cdots & * \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & a_{sj_s} & * & \cdots & * \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \end{pmatrix}$$

A matrix is said to be **reduced row-echelon form** if it is in row-echelon form and has the following additional properties:

3. The **leading entry** of every nonzero row is 1.
4. Each column that contains a leading entry has zeros everywhere else.

$$\begin{pmatrix} 0 & \cdots & 0 & \boxed{1} & * & \cdots & * & 0 & * & \cdots & * & 0 & * & \cdots & * & 0 & * & \cdots & * \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & \boxed{1} & * & \cdots & * & 0 & * & \cdots & * & 0 & * & \cdots & * \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & \boxed{1} & * & \cdots & * & 0 & * & \cdots & * \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & \boxed{1} & * & \cdots & * \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \\ \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 \end{pmatrix}$$

Remark 1.3:

1. In some text books, the row-echelon form is required to have the additional property 3.
2. Every matrix can be reduced to a row-echelon matrix (a matrix in row-echelon form) by some elementary row operations.
3. Every matrix has a unique reduced row-echelon form but it can have many different row-echelon forms.
4. The zero matrix 0 (the matrix with all entries are zeros) is a row echelon matrix and also row reduced echelon matrix.

Example 1.6:

1. The following augmented matrices are in reduced row-echelon form.

$$\left(\begin{array}{cc|c} 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right) \quad \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right) \quad \left(\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right) \quad \left(\begin{array}{cccc|c} 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

2. The following augmented matrices are in row-echelon form but not in reduced row-echelon form.

$$\left(\begin{array}{cc|c} 1 & -1 & 0 \\ 0 & 1 & 0 \end{array}\right) \quad \left(\begin{array}{ccc|c} -1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 2 & 3 \end{array}\right) \quad \left(\begin{array}{cc|c} 2 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array}\right) \quad \left(\begin{array}{cccc|c} 0 & 1 & 2 & 8 & 1 \\ 0 & 0 & 0 & 4 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

3. The following augmented matrices are not in row-echelon form. Hence, the matrices are also not in reduced row-echelon form.

$$\left(\begin{array}{cc|c} 0 & 1 & 0 \\ 1 & 0 & 0 \end{array}\right) \quad \left(\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & 1 & 3 \end{array}\right) \quad \left(\begin{array}{cc|c} 0 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 0 & 1 \end{array}\right) \quad \left(\begin{array}{cccc|c} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

Remark 1.4:

When the augmented matrix of a system of linear equations is in row-echelon form or reduced row-echelon form, we can get the solutions of the system easily.

Example 1.8

1. The array $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & -1 & 0 & 3 \\ 0 & 0 & 2 & 4 \end{array}\right]$ is the augmented matrix of a system of linear equations in 3 variables x_1, x_2, x_3 :

$$\begin{aligned} x_1 &= 2 \\ -x_2 &= 3 \\ 2x_3 &= 4. \end{aligned}$$

The system has only one solution: $x_1 = 2, x_2 = -3, x_3 = 2$.

2. The array $\left[\begin{array}{ccccc|c} 0 & 2 & 2 & 1 & -2 & 2 \\ 0 & 0 & 1 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 & 2 & 4 \end{array}\right]$ is the augmented matrix of a system of linear equations in 5 variables x_1, x_2, x_3, x_4, x_5 :

$$\begin{aligned} 2x_2 + 2x_3 + x_4 - 2x_5 &= 2 \\ x_3 + x_4 + x_5 &= 3 \\ 2x_5 &= 4. \end{aligned}$$

Note that x_1 does not appear in the equations and this means x_1 is arbitrary.

By the third equation $x_5 = 2$. Substitute $x_5 = 2$ into the second equation, we have $x_3 = 1 - x_4$.

Substitute $x_3 = 1 - x_4$ and $x_5 = 2$ into the first equation, we have $x_2 = 2 + \frac{1}{2}x_4$. (This method is called the **back-substitution**.)

Let $x_1 = s$ and $x_4 = t$ where s, t are arbitrary parameters.

$$x_1 = s$$

$$x_2 = 2 + \frac{1}{2}t$$

Then, the general solution is $x_3 = 1 - t$, $s, t \in \mathbb{R}$,

$$x_4 = t$$

$$x_5 = 2$$

and this system has infinitely many solutions.

3. The array $\left[\begin{array}{cccc|c} 1 & -1 & 0 & 3 & -2 \\ 0 & 0 & 1 & 2 & 5 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$ is the augmented matrix of a system of linear equations in

4 variables x_1, x_2, x_3, x_4 :

$$\begin{aligned} x_1 - x_2 + 3x_4 &= -2 \\ x_3 + 2x_4 &= 5 \\ 0 &= 0. \end{aligned}$$

Let $x_2 = s$ and $x_4 = t$ where s, t are arbitrary parameters.

$$x_1 = -2 + s - 3t$$

Then, the general solution is $x_2 = s$, $x_3 = 5 - 2t$, $x_4 = t$, $s, t \in \mathbb{R}$

and this system has infinitely many solutions.

4. The array $\left[\begin{array}{cc|c} 3 & 1 & 4 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{array} \right]$ is the augmented matrix of a system of linear equations in 2 variables

x_1, x_2 :

$$\begin{aligned} 3x_1 + x_2 &= 4 \\ 2x_2 &= 1 \\ 0 &= 1. \end{aligned}$$

This system is inconsistent. That is, the system has no solution.

1.3 Gaussian Elimination

Algorithm 1.1 (Gaussian Elimination)

The following procedures reduce a matrix to a row echelon form by some elementary row operations.

Step 1: Locate the leftmost nonzero column (a column with at least one nonzero entries.)

Step 2: Interchange the top row with another row, if necessary, to bring a nonzero entry to the top of the column found in Step 1.

Step 3: For each row below the top row, apply some elementary row operations to the rows below so that the entries below the leading entry of the top row become zeros.

Step 4: Cover the top row in the matrix and apply Step 1 to Step 3 to the submatrix that remains. Continue in this way until the entire matrix is in row-echelon form.

Remark 1.5:

We may write a system of linear equations in its augmented matrix $[A \mid B]$, then apply Gauss elimination to reduce the augmented matrix to a row-echelon form, which we shall then readily obtain the values of the unknowns by “*back substitution*”.

Algorithm 1.2 (Gaussian-Jordan Elimination)

Given a matrix, use Gauss Elimination (Algorithm 1.1) to reduce it to a row-echelon form. Then follow the following procedures reduce it to reduced row-echelon form:

Step 5: Multiply a suitable constant to each row so that all leading entries become 1.

Step 6: Beginning with the last nonzero row and working upward, apply some elementary row operation $R_i \rightarrow R_i + \beta R_j$, $\beta \in \mathbb{R}$ to the row above to introduce zero above the leading entries.

Example 1.9:

Use Gaussian Elimination to reduce the matrix $\begin{bmatrix} 0 & 0 & 2 & 4 & 2 & 8 \\ 1 & 2 & 4 & 5 & 3 & -9 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{bmatrix}$ to a row-echelon form.

Solution:

Step 1: The first column is the leftmost nonzero column.

Step 2: Interchange the first and second rows.

$$\begin{bmatrix} 0 & 0 & 2 & 4 & 2 & 8 \\ 1 & 2 & 4 & 5 & 3 & -9 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{bmatrix}$$

Step 3: Use an elementary row operations to the third row so that the entry (3, 1) is zero

$$\xrightarrow{R_3 \rightarrow R_3 + 2R_1} \begin{bmatrix} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 3 & 6 & 9 & -12 \end{bmatrix}$$

Step 4: Cover the first row and apply Step 1 to Step 3 to the submatrix that remains.

Step 1.1: The third column is the leftmost nonzero column.

Step 2.1: No action is need.

Step 3.1: Use an elementary row operations to the third row so that the entry (3, 3) is zero.

$$\xrightarrow{R_3 \rightarrow R_3 - \frac{3}{2}R_2} \begin{bmatrix} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 0 & 0 & 6 & -24 \end{bmatrix}$$

Step 4.1: The matrix is already in row-echelon form.

Example 1.10:

Use Gaussian-Jordan Elimination to reduce the matrix in Example 1.9 to a reduced row-echelon form.

Solution:

First, we follow the steps in Example 4.9 to obtain a row-echelon form as shown in Step 3.1.

Step 5: Multiply a suitable constant to each row so that all leading entries (the entries (1, 1), (2, 3), and (3, 5)) become 1.

$$\begin{array}{c} R_2 \rightarrow \frac{1}{2}R_2 \\ R_3 \rightarrow \frac{1}{6}R_3 \end{array} \rightarrow \begin{bmatrix} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 1 & 2 & 1 & 4 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{bmatrix}$$

Step 6: Apply some elementary row operations to the first and second rows to introduce zero to the entries above the leading entry of the third row. That is, introduce zero to the entries (1, 5) and (2, 5).

$$\begin{array}{c} R_1 \rightarrow R_1 - 3R_3 \\ R_2 \rightarrow R_2 - R_3 \end{array} \rightarrow \begin{bmatrix} 1 & 2 & 4 & 5 & 0 & 3 \\ 0 & 0 & 1 & 2 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{bmatrix}$$

Repeat Step 6, apply an elementary row operation to the first row to introduce zero to the entry above the leading entry of the second row. That is, introduce zero to the entries (1, 3).

$$\begin{array}{c} R_1 \rightarrow R_1 - 4R_2 \end{array} \rightarrow \begin{bmatrix} 1 & 2 & 0 & -3 & 0 & -29 \\ 0 & 0 & 1 & 2 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{bmatrix}.$$

The matrix is now in reduced row-echelon form.

Example 1.11:

Solve the system of linear equations

$$\begin{aligned} 2x_3 + 4x_4 + 2x_5 &= 8 \\ x_1 + 2x_2 + 4x_3 + 5x_4 + 3x_5 &= -9 \\ -2x_1 - 4x_2 - 5x_3 - 4x_4 + 3x_5 &= 6. \end{aligned}$$

Solution:

The augmented matrix of a system of linear equations is $\left[\begin{array}{ccccc|c} 0 & 0 & 2 & 4 & 2 & 8 \\ 1 & 2 & 4 & 5 & 3 & -9 \\ -2 & -4 & -5 & -4 & 3 & 6 \end{array} \right]$.

Method 1: Use Gaussian Elimination to reduce the augmented matrix to row-echelon form.

From Example 1.9, a row-echelon form of the above augmented matrix is

$$\left[\begin{array}{ccccc|c} 1 & 2 & 4 & 5 & 3 & -9 \\ 0 & 0 & 2 & 4 & 2 & 8 \\ 0 & 0 & 0 & 0 & 6 & -24 \end{array} \right]$$

which corresponds to the system:

$$\begin{aligned} x_1 + 2x_2 + 4x_3 + 5x_4 + 3x_5 &= -9 \\ 2x_3 + 4x_4 + 2x_5 &= 8 \\ 6x_5 &= -24. \end{aligned}$$

By the method of back-substitution, we have the third equation, $x_5 = -4$.

Substitute $x_5 = -4$ into the second equation, we have $2x_3 + 4x_4 + 2(-4) = 8 \Leftrightarrow x_3 = 8 - 2x_4$.

Substitute $x_5 = -4$ and $x_3 = 8 - 2x_4$ into the first equation, we have

$$x_1 + 2x_2 + 4(8 - 2x_4) + 5x_4 + 3(-4) = -9 \Leftrightarrow x_1 = -29 - 2x_2 + 3x_4.$$

Let $x_2 = s$ and $x_3 = t$ where s, t are arbitrary parameters.

$$x_1 = -29 - 2s + 3t$$

$$x_2 = s$$

Then, the general solution is $x_3 = 8 - 2t$, $s, t \in \mathbb{R}$.

$$x_4 = t$$

$$x_5 = -4$$

Method 2: Use Gaussian-Jordan Elimination to reduce the augmented matrix to reduced row-echelon form. From Example 1.10, the reduced row-echelon form of the augmented matrix is

$$\left[\begin{array}{ccccc|c} 1 & 2 & 0 & -3 & 0 & -29 \\ 0 & 0 & 1 & 2 & 0 & 8 \\ 0 & 0 & 0 & 0 & 1 & -4 \end{array} \right].$$

which corresponds to the system:

$$\begin{aligned} x_1 + 2x_2 + \quad - 3x_4 &= -29 \\ \quad \quad \quad x_3 + 2x_4 &= 8 \\ \quad \quad \quad \quad \quad x_5 &= -4. \end{aligned}$$

By the method of back-substitution, we have $x_5 = -4$, $x_3 = 8 - 2x_4$ and $x_1 = -29 - 2x_2 + 3x_4$.

Let $x_2 = s$ and $x_3 = t$ where s, t are arbitrary parameters.

$$x_1 = -29 - 2s + 3t$$

$$x_2 = s$$

Then, the general solution is $x_3 = 8 - 2t$, $s, t \in \mathbb{R}$.

$$x_4 = t$$

$$x_5 = -4$$

Example 1.12:

Determine whether the following system of linear equations is consistent or inconsistent:

$$\begin{aligned} 3x_1 + 2x_2 + x_3 &= 3 \\ 2x_1 + x_2 + x_3 &= 0 \\ 6x_1 + 2x_2 + 4x_3 &= 6. \end{aligned}$$

Solution:

The augmented matrix of the system of linear equations is

$$\left[\begin{array}{ccc|c} 3 & 2 & 1 & 3 \\ 2 & 1 & 1 & 0 \\ 6 & 2 & 4 & 6 \end{array} \right] \xrightarrow{\substack{R_2 \rightarrow R_2 - \frac{2}{3}R_1 \\ R_3 \rightarrow R_3 - 2R_1}} \left[\begin{array}{ccc|c} 3 & 2 & 1 & 3 \\ 0 & -\frac{1}{3} & -\frac{1}{3} & -2 \\ 0 & -2 & 2 & 0 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - 6R_2} \left[\begin{array}{ccc|c} 3 & 2 & 1 & 3 \\ 0 & -\frac{1}{3} & -\frac{1}{3} & -2 \\ 0 & 0 & 0 & 12 \end{array} \right]$$

From the third row, we have $0(x_3) = 12$.

However, $0 \neq 12$. Therefore, the system has no solution.

Hence, the system is inconsistent.

Alternative method: (To avoid fraction)

$$\begin{bmatrix} 3 & 2 & 1 & 3 \\ 2 & 1 & 1 & 0 \\ 6 & 2 & 4 & 6 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2} \begin{bmatrix} 1 & 1 & 0 & 3 \\ 2 & 1 & 1 & 0 \\ 6 & 2 & 4 & 6 \end{bmatrix} \xrightarrow{\begin{matrix} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 6R_1 \end{matrix}} \begin{bmatrix} 1 & 1 & 0 & 3 \\ 0 & -1 & 1 & 6 \\ 0 & -4 & 4 & 12 \end{bmatrix}$$

$$\xrightarrow{R_3 \rightarrow R_3 - 4R_2} \begin{bmatrix} 1 & 1 & 0 & 3 \\ 0 & -1 & 1 & 6 \\ 0 & 0 & 0 & -12 \end{bmatrix}.$$

From the third row, we have $0(x_3) = -12$.

However, $0 \neq -12$. Therefore, the system has no solution.

Hence, the system is inconsistent.

Remark 1.6:

1. A system of linear equations is inconsistent (that is it has no solution), if a row-echelon form of its augmented matrix has a row with nonzero last entry but zero elsewhere:

where $\otimes$ represents the leading entries and $\star$ is a nonzero number.

For example, $\left[\begin{array}{ccc|c} 1 & 2 & 4 & 2 \\ 0 & -1 & 0 & 3 \\ 0 & 0 & 0 & 4 \end{array} \right]$ is a row-echelon form of the augmented matrix of an inconsistent system.

2. A system of linear equations has exactly one (unique) solution, if a row-echelon form of its augmented matrix has a leading entry in every column, except the last column:

where $\otimes$ represents the leading entries.

In other words, a consistent linear system has exactly one solution if the number of variables in the linear system is equal to the number of nonzero rows in the row-echelon form of its augmented matrix.

For example, $\left[\begin{array}{ccc|c} 1 & 2 & 4 & 2 \\ 0 & -1 & 0 & 3 \\ 0 & 0 & 2 & 4 \end{array} \right]$ and $\left[\begin{array}{cccc|c} 1 & 1 & 2 & 3 & 4 \\ 0 & 2 & 0 & 1 & -1 \\ 0 & 0 & 4 & -1 & 2 \\ 0 & 0 & 0 & -1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right]$ are augmented matrix of some

consistent systems with exactly one solution.

3. A system of linear equations has infinitely many solution, if a row-echelon form of its augmented matrix has a column which is not the last column and does not consist of leading entry:

a column (not the last column) with no leading entry

where $\otimes$ represents the leading entries.

In other words, a consistent linear system has infinitely many solution if the number of variables in the linear system is greater than the number of nonzero rows in the row-echelon form of its augmented matrix.

For example, $\left[\begin{array}{cccc|c} 1 & -1 & 0 & 3 & 0 \\ 0 & 0 & 1 & 2 & 5 \\ 0 & 0 & 0 & 1 & 3 \end{array} \right]$ and $\left[\begin{array}{cccc|c} 1 & 2 & 0 & 3 & 0 \\ 0 & -1 & 0 & 0 & 5 \\ 0 & 0 & 0 & 1 & 3 \end{array} \right]$ are augmented matrix of some consistent systems with infinitely many solution.

Example 1.13:

What is the condition that must be satisfied by a , b and c so that the following system of linear equations is consistent?

$$\begin{aligned} x + 2y - 3z &= a \\ 2x + 6y - 11z &= b \\ x - 2y + 7z &= c \end{aligned}$$

Solution:

The augmented matrix of the system of linear equations is

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & a \\ 2 & 6 & -11 & b \\ 1 & -2 & 7 & c \end{array} \right] \xrightarrow{\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1}} \left[\begin{array}{ccc|c} 1 & 2 & -3 & a \\ 0 & 2 & -5 & b - 2a \\ 0 & -4 & 10 & c - a \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 + 2R_2} \left[\begin{array}{ccc|c} 1 & 2 & -3 & a \\ 0 & 2 & -5 & b - 2a \\ 0 & 0 & 0 & 2b + c - 5a \end{array} \right].$$

$\therefore$ The system is consistent if and only if $2b + c - 5a = 0$.

Example 1.14:

Determine the values of k so that the following system of linear equations

$$\begin{aligned} x + 2y + z &= 1 \\ 2x + ky + 2z &= 2 \\ 4x + 8y + k^2z &= 2k \end{aligned}$$

has (a) no solution (b) a unique solution (c) infinitely many solutions.

Solution:

The augmented matrix of the system of linear equations is

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & k & 2 & 2 \\ 4 & 8 & k^2 & 2k \end{array} \right] \xrightarrow{\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 4R_1}} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & k-4 & 0 & 0 \\ 0 & 0 & k^2-4 & 2k-4 \end{array} \right].$$

- (a) The system has no solution when $k^2 - 4 = 0$ and $2k - 4 \neq 0$. That is, $k = -2$.
- (b) The system has a unique solution when $k - 4 \neq 0$ and $k^2 - 4 \neq 0$. That is, $k \neq \pm 2$ and $k \neq 4$.
- (c) The system has infinitely many solutions when $k - 4 = 0$ or both $k^2 - 4 = 0$ and $2k - 4 = 0$. That is, $k = 2$ or $k = 4$.

1.4 Homogeneous system ($B = 0$)

Definition 1.7. A system of m linear equations in n unknowns $x_1, x_2, \dots, x_n$ is said to be **homogeneous** if it has the form

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= 0 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= 0 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= 0 \end{aligned}$$

where $a_{ij}, 1 \leq i \leq m, 1 \leq j \leq n$ are real constants.

A homogeneous system always has a solution, $x_1 = 0, x_2 = 0, x_3 = 0, \dots, x_n = 0$. This solution is called **the trivial solution**. Any other solution, if it exists, is called **nontrivial solution**.

Remark 1.8:

1. A homogeneous system of linear equations is consistent, it has either only the trivial solution or infinitely many solutions in addition to the trivial solution.
2. A homogeneous system of linear equations with more unknowns than equations has infinitely many solutions.

Example 1.15: Determine the value(s) of k such that

$$\begin{aligned} x - 5y + 3z &= 0 \\ 5x - y - kz &= 0 \\ x - 2y + kz &= 0 \end{aligned}$$

has nontrivial solution.

Solution:

The augmented matrix of the system of linear equations is

$$\left[\begin{array}{ccc|c} 1 & -5 & 3 & 0 \\ 5 & -1 & -k & 0 \\ 1 & -2 & k & 0 \end{array} \right] \xrightarrow{\substack{R_2 \rightarrow R_2 - 5R_1 \\ R_3 \rightarrow R_3 - R_1}} \left[\begin{array}{ccc|c} 1 & -5 & 3 & 0 \\ 0 & 24 & -k-15 & 0 \\ 0 & 3 & k-3 & 0 \end{array} \right] \xrightarrow{R_3 \rightarrow 8R_3} \left[\begin{array}{ccc|c} 1 & -5 & 3 & 0 \\ 0 & 24 & -k-15 & 0 \\ 0 & 24 & 8k-24 & 0 \end{array} \right] \xrightarrow{R_3 \rightarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & -5 & 3 & 0 \\ 0 & 24 & -k-15 & 0 \\ 0 & 0 & 9k-9 & 0 \end{array} \right]$$

$\therefore$ The system has nontrivial solution if $9k - 9 = 0$, that is $k = 1$.

(Remark: If $k \neq 1$, then the system has only trivial solution $x = y = z = 0$.)

Chapter 2: Inverses of Square Matrices

2.1 Introduction to Matrices

Definition 2.1: Matrix

Let F be a real field or complex field. An $m \times n$ (m by n) **matrix** (plural **matrices**) over F is an array of numbers arranged in m rows and n columns.

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix} = [a_{ij}]_{m \times n}$$

When the size of the matrix is known, we may just write $A = [a_{ij}]$.

The a_{ij} is called **(i, j)-entry** of A , which is the number in the i -th row and j -th column of the matrix A . The totality of all $m \times n$ matrices A over F is denoted by $M_{m,n}(F)$, or simply $M_{m,n}$. For short, we shall write $M_n(F)$ for $M_{n,n}$.

Definition 2.2: Column matrix and row matrix

A **column matrix** (or **column vector**) is a matrix with only one column, that is, it is an $m \times 1$ matrix.

A **row matrix** (or **row vector**) is a matrix with only one row, that is, it is an $1 \times n$ matrix.

Definition 2.3: Square matrices

A matrix is called a **square matrix** if it has the same number of rows and columns. An $n \times n$ matrix is called a **square matrix of order n** or **n -square matrix**.

The **diagonal** (or **main diagonal**) of a n -square matrix A is the sequence of entries $a_{11}, a_{22}, \dots, a_{nn}$. The entries a_{ii} are called the **diagonal entries** of A , while $a_{ij}, i \neq j$, are called the **off-diagonal** or **nondiagonal entries** of A .

Example 2.1:

$$\begin{bmatrix} -2 & 1 & -1 & 0 \\ 0 & 3 & 1 & 2 \end{bmatrix} \in M_{2,4}(\mathbb{R}) \text{ is a } 2 \times 4 \text{ matrix;}$$

$$[3 \quad 1+i \quad i \quad -5 \quad 7] \in M_{1,5}(\mathbb{C}) \text{ is a } 1 \times 5 \text{ row matrix;}$$

$$\begin{bmatrix} 1 \\ 3 \\ 0 \\ -4 \end{bmatrix} \in M_{4,1}(\mathbb{R}) \text{ is a } 4 \times 1 \text{ column matrix;}$$

$$\begin{bmatrix} 1 & 2 & 4 \\ -1 & 0 & 3 \\ 3 & 1 & -2 \end{bmatrix} \in M_3(\mathbb{R}) \text{ is a 3-square matrix.}$$

Definition 2.4: Some special square matrices**(1) Diagonal Matrix**

A **diagonal matrix** is a square matrix $A = [a_{ij}]$ whose non-diagonal entries are all zero. That is, $a_{ij} = 0 \quad \forall i \neq j$. It is denoted by $A = \text{diag}(a_{11}, a_{22}, \dots, a_{nn})$.

(2) Scalar matrix

A **scalar matrix** is a diagonal matrix with all its diagonal entries are the same. That is, for a constant c , $a_{ii} = c \quad \forall i = 1, 2, \dots, n$ and $a_{ij} = 0 \quad \forall i \neq j$.

(3) Identity matrix I_n

An **identity matrix** (or **unit matrix**) is a diagonal matrix with all its diagonal entries are 1, and it is denoted by I_n . That is, $a_{ii} = 1 \quad \forall i = 1, 2, \dots, n$ and $a_{ij} = 0 \quad \forall i \neq j$.

(4) Zero matrix $0_{m \times n}$

A **zero matrix** is an $m \times n$ matrix with all entries equal to zero, and it is denoted by $0_{m \times n}$, or simply 0 . That is, $a_{ij} = 0 \quad \forall i = 1, 2, \dots, m$ and $j = 1, 2, \dots, n$.

(5) Triangular matrices

An **upper triangular matrix** is a square matrix $A = [a_{ij}]$ whose entries below the main diagonal are all zero. That is, $a_{ij} = 0 \quad \forall i > j$.

A **lower triangular matrix** is a square matrix $A = [a_{ij}]$ whose entries above the main diagonal are all zero. That is, $a_{ij} = 0 \quad \forall i < j$.

Both upper and lower triangular matrices are called triangular matrices.

(6) Symmetric matrix

A **symmetric matrix** is a square matrix $A = [a_{ij}]$ whose transpose equals the matrix ($A = A^T$).

That is, $a_{ij} = a_{ji} \quad \forall i, j$.

Example 2.2:

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 & -8 \\ 0 & -2 & 5 \\ 0 & 0 & -7 \end{bmatrix},$$

$$C = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 6 & 1 & 0 & 0 \\ 5 & -2 & -8 & 0 \\ 0 & 7 & 0 & 4 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 3 & -1 \\ 3 & -2 & 0 \\ -1 & 0 & 7 \end{bmatrix}, \quad E = \begin{bmatrix} -1 & 6 & 1 & 0 \\ 6 & 0 & 2 & 7 \\ 1 & 2 & -8 & -3 \\ 0 & 7 & -3 & 4 \end{bmatrix}.$$

I_3 and I_4 are identity matrices of order 3 and 4, respectively.

A the diagonal matrix $\text{diag}(2, -1, 0, 5)$; B is an upper triangular matrix; C is a lower triangular matrix; D and E are symmetric matrices

Definition 2.5: Equal matrices

Two matrices $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{p \times q}$ are said to be equal if they have the same size and their corresponding entries are equal. That is, $m = p$, $n = q$ and $a_{ij} = b_{ij}$ for all i and j .

2.2: Matrix Operations

Definition 2.6: Matrix Addition, Subtraction and Scalar Multiplication

Let $A = [a_{ij}]_{m \times n}$, $B = [b_{ij}]_{m \times n}$ and $C = [c_{ij}]_{m \times n}$ be three $m \times n$ matrices.

1. Matrix Addition: $A + B = C \Leftrightarrow c_{ij} = a_{ij} + b_{ij}.$
2. Matrix Subtraction: $A - B = C \Leftrightarrow c_{ij} = a_{ij} - b_{ij}.$
3. Scalar Multiplication: $\lambda A = C \Leftrightarrow c_{ij} = \lambda a_{ij}$ for any scalar $\lambda \in F$.

Remark 21.:

1. Given a matrix A , $-A$ is used to denote $(-1)A$.
2. Given two matrices A and B of the same size, $A - B$ is defined to be $A + (-B)$.

Definition 2.7: Matrix Multiplication

If $A = [a_{ij}]_{m \times r}$ is an $m \times r$ matrix and $B = [b_{ij}]_{r \times n}$ is an $r \times n$ matrix, then the matrix product AB is the $m \times n$ matrix whose (i, j) -entry is given by

$$a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{ir}b_{rj} = \sum_{k=1}^r a_{ik}b_{kj}, \quad i = 1, 2, \dots, m, \quad j = 1, 2, \dots, n.$$

Example 2.3:

$$\text{Let } A = \begin{bmatrix} -2 & 1 & -1 & 0 \\ 0 & 3 & 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 4 & 1 & 6 & 2 \end{bmatrix} \text{ and } C = \begin{bmatrix} -1 & 8 \\ 2 & 1 \\ 1 & 1 \\ 12 & 6 \end{bmatrix}$$

$$\text{Then } A + B = \begin{bmatrix} -2+1 & 1+1 & -1+2 & 0+1 \\ 0+4 & 3+1 & 1+6 & 2+2 \end{bmatrix} = \begin{bmatrix} -1 & 2 & 1 & 1 \\ 4 & 4 & 7 & 4 \end{bmatrix}, \quad A - B = \begin{bmatrix} -3 & 0 & -3 & -1 \\ -4 & 2 & -5 & 0 \end{bmatrix},$$

$$BC = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 4 & 1 & 6 & 2 \end{bmatrix} \begin{bmatrix} -1 & 8 \\ 2 & 1 \\ 1 & 1 \\ 12 & 6 \end{bmatrix} = \begin{bmatrix} 1(-1)+1(2)+2(1)+1(12) & 1(8)+1(1)+2(1)+1(6) \\ 4(-1)+1(2)+6(1)+2(12) & 4(8)+1(1)+6(1)+2(6) \end{bmatrix} = \begin{bmatrix} 15 & 17 \\ 28 & 51 \end{bmatrix},$$

$$CB = \begin{bmatrix} -1 & 8 \\ 2 & 1 \\ 1 & 1 \\ 12 & 6 \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 & 1 \\ 4 & 1 & 6 & 2 \end{bmatrix} = \begin{bmatrix} 31 & 7 & 46 & 15 \\ 6 & 3 & 10 & 4 \\ 5 & 2 & 8 & 3 \\ 36 & 18 & 60 & 24 \end{bmatrix} \text{ and } -3C = \begin{bmatrix} 3 & -24 \\ -6 & -3 \\ -3 & -3 \\ -36 & -18 \end{bmatrix}.$$

Properties of the addition of matrices

Let $A, B, C \in M_{m \times n}(F)$ be three $m \times n$ matrices. Then,

- (1) **Commutative Law:** $A + B = B + A.$
- (2) **Associative Law:** $(A + B) + C = A + (B + C).$
- (3) **Distributive law:** $\lambda(A + B) = \lambda A + \lambda B$ and $(\lambda + \beta)A = \lambda A + \beta A$ for any constants λ and β .

Properties of the multiplication of matrices

Let $A, A_1, A_2 \in M_{m \times n}(F)$, $B, B_1, B_2 \in M_{n \times p}(F)$ and $C \in M_{p \times q}(F)$. Then,

- (1) **Associative Law:** $(AB)C = A(BC)$.
- (2) **Distributive law:** $\lambda(AB) = (\lambda A)B = A(\lambda B)$ for any constant λ ,
 $A(B_1 + B_2) = AB_1 + AB_2$,
 $(A_1 + A_2)B = A_1B + A_2B$.
- (3) **Identity law:** $AI_n = A = I_n$.

Remark:

1. The multiplication of matrices **does not satisfy commutative law**. That is, in general, $AB \neq BA$. (See Example 2.3.) To distinguish AB and BA , we refer to AB as the **pre-multiplication** of A to B and BA as the **post-multiplication** of A to B .
2. $AB = 0$ **does not imply** $A = 0$ or $B = 0$ or $BA = 0$.

Example 2.4: There exists matrices A and B such that $AB = 0$ but $A \neq 0$, $B \neq 0$ and $BA \neq 0$,

$$AB = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0 \text{ but } A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \neq 0, B = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \neq 0 \text{ and}$$

$$BA = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ -2 & -2 \end{bmatrix} \neq 0.$$

Definition 2.8: Transpose

The **transpose** of a matrix $A = [a_{ij}]_{m \times n}$, written $A^T = [a_{ji}]_{n \times m}$ (or A^t), is the matrix obtained from A by changing columns to rows (and rows to columns).

That is, if $A = [a_{ij}] \in M_{mn}$, then $A^T = [b_{ij}] \in M_{nm}$ where $b_{ij} = a_{ji}$, $i = 1, 2, \dots, n$, $j = 1, 2, \dots, m$.

Example 2.5: $A = \begin{bmatrix} 1 & 0 & -8 \\ 3 & -2 & 5 \\ 6 & 9 & -7 \end{bmatrix}$ and $A^T = \begin{bmatrix} 1 & 3 & 6 \\ 0 & -2 & 9 \\ -8 & 5 & -7 \end{bmatrix}$; $B = \begin{bmatrix} 2 & 3 \\ 1 & 2 \\ 4 & 5 \end{bmatrix}$ and $B^T = \begin{bmatrix} 2 & 1 & 4 \\ 3 & 2 & 5 \end{bmatrix}$.

Properties of transpose of matrix

Let $A = [a_{ij}]_{m \times n}$ be an $m \times n$ matrix. Then

- (1) $(A^T)^T = A$.
- (2) $(A + B)^T = A^T + B^T$ if $B = [b_{ij}]_{m \times n}$ is an $m \times n$ matrix.
- (3) $(\lambda A)^T = \lambda A^T$ for a constant λ .
- (4) $(AB)^T = B^T A^T$ if $B = [b_{ij}]_{n \times p}$ is an $n \times p$ matrix.

Proof (Property (4)):

Let $A = [a_{ij}]_{m \times n}$ and $B = [b_{ij}]_{n \times p}$, where $i = 1, 2, \dots, m$, $j = 1, 2, \dots, n$.

- (i) Since the size of AB is $m \times p$, then the size of $(AB)^T$ is $p \times m$.
On the other hand, since the size of B^T is $p \times n$ and the size of A^T is $n \times m$, then the size of $B^T A^T$ is $p \times m$. Thus, $(AB)^T$ and $B^T A^T$ have the same size.
- (ii) Since the (i, j) -entry of AB is $a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{in}b_{nj}$, then the (i, j) -entry of $(AB)^T$ is $a_{j1}b_{1i} + a_{j2}b_{2i} + \dots + a_{jn}b_{ni}$.
On the other hand, let $A^T = [a'_{ij}]$ and $B^T = [b'_{ij}]$, that is, $a'_{ij} = a_{ji}$ and $b'_{ij} = b_{ji}$. Then, the (j, i) -entry of the matrix $B^T A^T$ is given by $b'_{i1}a'_{1j} + b'_{i2}a'_{2j} + \dots + b'_{in}a'_{nj}$
 $= b_{1i}a_{j1} + b_{2i}a_{j2} + \dots + b_{ni}a_{jn} = a_{j1}b_{1i} + a_{j2}b_{2i} + \dots + a_{jn}b_{ni} =$ the (i, j) -entry of $(AB)^T$.
By (i) and (ii), we have $B^T A^T = (AB)^T$.

2.3 Inverse of a Matrix

Definition 2.9: Invertible matrix

An n -square matrix $A \in M_n$ is said to be **invertible** if there exists a n -square matrix $B \in M_n$ such that $AB = I_n$ and $BA = I_n$. Such a matrix B is called an **inverse of A** .

If a matrix A has an inverse, then A is also called a **nonsingular matrix**. If A has no inverse, then A is called a **singular matrix**.

Example 2.6:

1. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$. Then

$$AB = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix} \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3 \text{ and}$$

$$BA = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3.$$

Hence, A is invertible and B is an inverse of A . B is also invertible and A is an inverse of B .

That is $A^{-1} = B$ and $B^{-1} = A$.

2. Consider the matrix equation $\begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} X = \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix}$.

By pre-multiplying the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$, which is the inverse of $\begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$, to both

sides of the matrix equation, we have

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix} \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} X = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 0 \end{bmatrix} \Rightarrow I_3 X = \begin{bmatrix} 6 \\ 13 \\ 4 \end{bmatrix} \Rightarrow X = \begin{bmatrix} 6 \\ 13 \\ 4 \end{bmatrix}.$$

3. Show that $A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$ is singular.

Solution: (Proof by contradiction)

Suppose on the contrary, A has an inverse $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. By the definition of inverse, we have

$$BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \text{ However, } BA = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} a+b & 0 \\ c+d & 0 \end{bmatrix} \neq I_2, \text{ which is a contradiction.}$$

Hence, A is singular.

Theorem 2.1: (Uniqueness of Inverses)

Let A be an n -square matrix. If A is invertible, then it has a unique inverse. That is, if B and C are inverses of A , then $B = C$.

Proof: Suppose B and C are inverses of A .

$$\begin{aligned} \text{Then } AB = I_n &\Rightarrow C(AB) = C I_n \Rightarrow (CA)B = C \Rightarrow I_n B = C \quad (C \text{ are inverses of } A \Rightarrow CA = I_n) \\ &\Rightarrow B = C. \end{aligned}$$

Remark 2.2:

1. **Notation of inverse:** Let A be an invertible matrix. By Theorem 2.1, there is only one (unique) inverse of A , and A^{-1} is used to denote this unique inverse of A .
2. **Cancellation Law for matrices:** Let A be an invertible m -square matrix.
 - (i) If B_1 and B_2 are $m \times n$ matrices with $AB_1 = AB_2$, then $B_1 = B_2$.
 - (ii) If C_1 and C_2 are $n \times m$ matrices with $C_1A = C_2A$, then $C_1 = C_2$.

Take note that, if the matrix A is not invertible, then the Cancellation Law may not hold.

For example, take $A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$, $B_1 = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$ and $B_2 = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$. Then $AB_1 = \begin{bmatrix} 2 & 0 \\ 2 & 0 \end{bmatrix} = AB_2$ but $B_1 \neq B_2$.

Example 2.7:

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $ad-bc \neq 0$. Show that A is invertible and $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

Solution:

Let $B = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix}$. Note that B is well-defined if $ad-bc \neq 0$.

$$AB = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2 \text{ and } BA = \begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2.$$

Thus, A is invertible and $A^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = B$.

Example 2.8: Let A be an n -square matrix such that $A^2 - 3A - 6I_n = 0_n$. Show that A is invertible.

Solution:

$$A^2 - 3A - 6I_n = 0_n \Rightarrow A^2 - 3A = 6I_n \Rightarrow A(A - 3I_n) = 6I_n \Rightarrow A\left(\frac{1}{6}(A - 3I_n)\right) = I_n.$$

and $\left(\frac{1}{6}(A - 3I_n)\right)A = \left(\frac{1}{6}(A^2 - 3A)\right) = I_n$. Hence, A is invertible and $A^{-1} = \frac{1}{6}(A - 3I_n)$.

Properties of inverse of a matrix:

Let $A = [a_{ij}]_n \in M_n$ and $B = [b_{ij}]_n \in M_n$ be n -square invertible matrices. Then the following properties hold.

1. λA is invertible and $(\lambda A)^{-1} = \frac{1}{\lambda} A^{-1}$ for a nonzero scalar λ .
2. A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.
3. A^{-1} is invertible, and $(A^{-1})^{-1} = A$.
4. AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

Proof: (Property 2)

To show that A^T is invertible, we only need to verify that $(A^{-1})^T$ is the inverse of A^T .

Note that $A^T(A^{-1})^T = (A^{-1}A)^T = (I_n)^T = I_n$ and $(A^{-1})^T A^T = (AA^{-1})^T = (I_n)^T = I_n$.

Thus, A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.

Remark 2.3:

By Property 4, if $A_1, A_2, \dots, A_k$ are invertible matrices of the same size, then $A_1 A_2 \cdots A_k$ is invertible and $(A_1 A_2 \cdots A_k)^{-1} = A_k^{-1} A_{k-1}^{-1} \cdots A_2^{-1} A_1^{-1}$.

Definition 2.10: Power of matrices

Let A be an n -square matrix and let k be a non-negative integer. We define A^k as follows:

$$A^k = \begin{cases} I_n & \text{if } k = 0 \\ \underbrace{AA \cdots A}_{k \text{ times}} & \text{if } k \geq 1. \end{cases}$$

If A is invertible, then for $k \geq 1$, we define $A^{-k} = (A^{-1})^k = \underbrace{A^{-1} A^{-1} \cdots A^{-1}}_{k \text{ times}}$.

Example 2.9: Let $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$.

Since $AB = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$ and $BA = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$,

then $A^{-1} = B$.

$$A^3 = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 11 & 30 \\ 15 & 41 \end{bmatrix},$$

$$A^{-3} = (A^{-1})^3 = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 41 & -30 \\ -15 & 11 \end{bmatrix}.$$

Remark 2.4:

1. Let A be an n -square matrix and let r and s be integers. Then $A^r A^s = A^{r+s}$.
2. If A is invertible, then A^k is invertible and $(A^k)^{-1} = A^{-k}$, where k is a non-negative integer.

2.4 Elementary Matrices

Definition 2.11: Elementary matrices

A n -square matrix $E \in M_n(\mathbb{R})$ is called an **elementary matrix** if E can be obtained from the identity matrix I_n by performing a single elementary row operation.

All **elementary matrices** are invertible and their inverses are also elementary matrices.

Consider the three types of Elementary Row Operations on Matrices (see Section 1.2).

1. Interchange the i th row and the j th row: $R_i \leftrightarrow R_j$.

Let A be an $m \times n$ matrix and let $E =$

$$E = \begin{pmatrix} 1 & & & & 0 \\ & \ddots & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \\ & & 0 & & 1 \\ & & 1 & & 0 \\ & & & & & \ddots & \\ 0 & & & & & & 1 \end{pmatrix}$$

$\uparrow$ i th column $\uparrow$ j th column

be an m -square matrix.

Then EA is an $m \times n$ matrix which can be obtained from A by interchanging the i -th and j -th row of A .

The matrix E is invertible and $E^{-1} = E$.

2. Multiplication of a row by a nonzero constant c : $R_i \rightarrow cR_i$, $c \in \mathbb{R} \setminus \{0\}$.

Let A be an $m \times n$ matrix and let $E =$

$$E = \begin{pmatrix} 1 & & & & 0 \\ & \ddots & & & \\ & & c & & \\ & & & \ddots & \\ & & & & 1 \\ & & 0 & & 1 \end{pmatrix}$$

$\uparrow$ i th column

be an m -square matrix.

Then EA is an $m \times n$ matrix which can be obtained from A by multiplying the i -th row by c .

Since $c \neq 0$, the matrix E is invertible and $E^{-1} =$

$$E^{-1} = \begin{pmatrix} 1 & & & & 0 \\ & \ddots & & & \\ & & \frac{1}{c} & & \\ & & & \ddots & \\ & & & & 1 \\ & & 0 & & 1 \end{pmatrix}$$

$\uparrow$ i th column

3. Addition of a constant multiple of one row to another: $R_j \rightarrow R_j + cR_i$, $c \in \mathbb{R}$.

Let A be an $m \times n$ matrix and let $E =$

$$\begin{pmatrix} 1 & & & & 0 \\ & \ddots & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \\ & & c & & 1 \\ & & & & \\ 0 & & & & & \ddots & \\ & & & & & & 1 \end{pmatrix} \quad \leftarrow j\text{th row} \quad \text{if } j > i$$

$\uparrow$ $\uparrow$
 $i\text{th column}$ $j\text{th column}$

or

$$\begin{pmatrix} 1 & & & & 0 \\ & \ddots & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \\ & & 1 & & c \\ & & & & \\ 0 & & & & & \ddots & \\ & & & & & & 1 \end{pmatrix} \quad \leftarrow j\text{th row} \quad \text{if } j < i.$$

$\uparrow$ $\uparrow$
 $j\text{th column}$ $i\text{th column}$

be an m -square matrix.

Then EA is an $m \times n$ matrix which can be obtained from A by adding c times the i -th row to the j -th row of A .

The matrix E is invertible and $E^{-1} =$

$$\begin{pmatrix} 1 & & & & 0 \\ & \ddots & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \\ & & -c & & 1 \\ & & & & \\ 0 & & & & & \ddots & \\ & & & & & & 1 \end{pmatrix} \quad \leftarrow j\text{th row} \quad \text{if } j > i$$

$\uparrow$ $\uparrow$
 $i\text{th column}$ $j\text{th column}$

or

$$\begin{pmatrix} 1 & & & & 0 \\ & \ddots & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \\ & & 1 & & -c \\ & & & & \\ 0 & & & & & \ddots & \\ & & & & & & 1 \end{pmatrix} \quad \leftarrow j\text{th row} \quad \text{if } j < i.$$

$\uparrow$ $\uparrow$
 $j\text{th column}$ $i\text{th column}$

Remark 2.5:

If a matrix B is obtained from A by performing by a finite sequence of elementary row operations, then there exist elementary matrices $E_1, E_2, \dots, E_k$ such that

$$B = E_k \cdots E_2 E_1 A \quad \text{and} \quad A = E_1^{-1} E_2^{-1} \cdots E_k^{-1} B.$$

Example 2.10:

Suppose the matrix B is obtained from A by performing elementary row operations as below.

For each of these elementary row operations, we write down the corresponding elementary matrices $E_k, k = 1, 2, 3, 4$.

$$\begin{aligned}
 A = \begin{bmatrix} 0 & -3 & 5 \\ 2 & 5 & 3 \\ 1 & 2 & 3 \end{bmatrix} &\xrightarrow[E_1]{R_1 \leftrightarrow R_3} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 0 & -3 & 5 \end{bmatrix} \xrightarrow[E_2]{R_2 \rightarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & -3 & 5 \end{bmatrix} \\
 &\xrightarrow[E_3]{R_3 \rightarrow R_3 + 3R_2} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & -4 \end{bmatrix} \xrightarrow[E_4]{R_3 \rightarrow -\frac{1}{4}R_3} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} = B,
 \end{aligned}$$

$E_1 A \qquad E_2(E_1 A) \qquad E_3(E_2 E_1 A) \qquad E_4(E_3 E_2 E_1 A)$

where $E_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$, $E_2 = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, $E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix}$, $E_4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1/4 \end{bmatrix}$.

Hence, $B = E_4 E_3 E_2 E_1 A$. That is,

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1/4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & -3 & 5 \\ 2 & 5 & 3 \\ 1 & 2 & 3 \end{bmatrix}.$$

For each $k = 1, 2, 3, 4$, E_k is invertible and

$$E_1^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \quad E_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad E_3^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}, \quad E_4^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -4 \end{bmatrix}.$$

Hence, $A = E_1^{-1} E_2^{-1} E_3^{-1} E_4^{-1} B$. That is,

$$A = \begin{bmatrix} 0 & -3 & 5 \\ 2 & 5 & 3 \\ 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -4 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix}.$$

Theorem 2.1:

If $A \in M_n(\mathbb{R})$, then the following statements are equivalent:

- (i) A is invertible.
- (ii) The homogenous linear system $AX = 0$ has only trivial solution.
- (iii) The linear system $AX = B$ has a unique solution for each $B \in M_{n,1}(\mathbb{R})$.
- (iv) The reduced row-echelon form of A is the identity matrix I_n .
- (v) The matrix A can be expressed as a product of elementary matrices.

Theorem 2.2: The Matrix Inversion Algorithm

Let $A \in M_n(\mathbb{R})$ be an n -square matrix. If A can be transformed to the identity matrix I_n by a finite sequence of elementary row operations $O_1, O_2, O_3, \dots, O_k$, then A is invertible, and the same finite sequence of elementary row operations applied to yields the inverse matrix A^{-1} , i.e.,

$$A \xrightarrow{O_1, O_2, \dots, O_k} I_n \Rightarrow I_n \xrightarrow{O_1, O_2, \dots, O_k} A$$

or $\left(A \mid I_n \right) \xrightarrow{O_1, O_2, \dots, O_k} \left(I_n \mid A^{-1} \right).$

Example 2.11:

Use the elementary row operations to determine the inverse of the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}.$$

Solution:

$$\begin{aligned} [A \mid I_3] &= \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1}} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -2 & 0 & 1 \end{array} \right] \\ &\xrightarrow{R_3 \rightarrow R_3 + 2R_2} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array} \right] \xrightarrow{R_3 \rightarrow -R_3} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \\ &\xrightarrow{\substack{R_1 \rightarrow R_1 - 3R_3 \\ R_2 \rightarrow R_2 + 3R_3}} \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & -14 & 6 & 3 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 - 2R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \\ &= [I_3 \mid A^{-1}]. \quad \therefore A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}. \end{aligned}$$

Remark 2.6:

By Theorem 2.1, a square matrix is invertible if and only if its reduced row-echelon form is an identity matrix. This can be used to check whether a square matrix is invertible. If the row-echelon form of a square matrix has no zero row, then the matrix is invertible; otherwise, the matrix is not invertible.

Example 2.12: Determine whether the matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 4 \\ 3 & 6 & 3 \end{bmatrix}$ is invertible or not.

Solution:

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 4 \\ 3 & 6 & 3 \end{bmatrix} \xrightarrow{\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1}} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 3 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - \frac{3}{4}R_2} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix}.$$

A is a square matrix and its row-echelon form has a zero row. Hence, A is not invertible.

2.5 Row Equivalence**Definition 2.12: Row Equivalence**

Let $A, B \in M_{m \times n}(\mathbb{R})$ be two $m \times n$ matrices. We say that A is **row equivalent** to B , denoted by $A \sim B$, if B can be obtained from A by performing a finite sequence of elementary row operations.

Example 2.12:

1. For any $m \times n$ matrix A , $A \sim A$. This is because $A \xrightarrow{R_1 \rightarrow 1R_1} A$.

2. Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 4 \\ 3 & 6 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 0 \end{bmatrix}$. From Example 2.11, B can be obtained from A by performing 3 elementary row operations. Hence, $A \sim B$.

3. The row equivalence is an equivalence relation on $M_{m \times n}(\mathbb{R})$. That is, let $A, B, C \in M_{m \times n}(\mathbb{R})$, then the following statements hold true.

(i) $A \sim A$,

(ii) $A \sim B \Rightarrow B \sim A$,

(iii) $A \sim B$ and $B \sim C \Rightarrow A \sim C$.

From (ii), both statements “ A is row equivalent to B ” and “ B is row equivalent to A ” can be replaced by the statement “ A and B are row equivalent”.

Theorem 2.3:

If $A, B \in M_{m \times n}(\mathbb{R})$, then the following statements are equivalent:

- (i) A is row equivalent to B .
- (ii) There exists elementary matrices $E_1, E_2, \dots, E_k \in M_m(\mathbb{R})$ such that $B = E_k E_{k-1} \cdots E_1 A$.
- (iii) There exists an invertible matrix $P \in M_m(\mathbb{R})$ such that $B = PA$.

Proof:

- (i) $\Rightarrow$ (ii): A is row equivalent to B means B can be obtained from A by performing a finite sequence of elementary row operations. Hence, there exists elementary matrices $E_1, E_2, \dots, E_k \in M_m(\mathbb{R})$ such that $B = E_k E_{k-1} \cdots E_1 A$.
- (ii) $\Rightarrow$ (iii): Let $P = E_k E_{k-1} \cdots E_1 \in M_m(\mathbb{R})$.
Since each $E_1, E_2, \dots, E_k$ is invertible, then $P = E_k E_{k-1} \cdots E_1$ is invertible.
Hence, there exists an invertible matrix $P \in M_m(\mathbb{R})$ such that $B = PA$.
- (iii) $\Rightarrow$ (i): If there exists an invertible matrix $P \in M_m(\mathbb{R})$ such that $B = PA$, then P can be expressed as product of elementary matrices $P = E_k E_{k-1} \cdots E_1$.
Hence, $B = PA = E_k E_{k-1} \cdots E_1 A$. Consequently, B can be obtained from A by performing a finite sequence of elementary row operations, which implies that A is row equivalent to B .

Theorem 3.3:

Let $A \in M_n(\mathbb{R})$. Then, A is invertible if and only if A and I_n are row equivalent.

Theorem 3.4:

Every matrix $A \in M_{m \times n}(\mathbb{R})$ is row equivalent to a row-echelon matrix.

Theorem 3.4:

Every matrix $A \in M_{m \times n}(\mathbb{R})$ has a unique reduced row-echelon matrix.

Chapter 3: Determinants and Rank

3.1 Determinants

Definition 3.1: Determinants

Let $A = [a_{ij}] \in M_n(\mathbb{R})$. The **determinant of A** , written as $\det A$ or $|A|$ is a scalar defined by

$$\det A = \begin{cases} a_{11} & \text{if } n = 1, \\ a_{11}A_{11} + a_{12}A_{12} + \dots + a_{1n}A_{1n} & \text{if } n \geq 2, \end{cases}$$

where, for each $1 \leq i, j \leq n$, let $A_{ij} = (-1)^{i+j} \det M_{ij}$ and let $M_{ij} \in M_{n-1}(\mathbb{R})$ be an $(n-1)$ -submatrix of A obtained by deleting the i -th row and the j -th column of A . The number A_{ij} is called the **cofactor of a_{ij} in $\det A$** and $\det M_{ij}$ is called the **minor of a_{ij} in $\det A$** .

This way we define $\det A$ is called **cofactor expansion**.

Notation: For a matrix $A = [a_{ij}] \in M_n(\mathbb{R})$, the determinant of A is denoted by

$$\begin{aligned} \det A = |A| &= \begin{vmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix} = a_{11} \begin{vmatrix} a_{22} & a_{23} & \cdots & a_{2n} \\ a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & & \vdots \\ a_{n2} & a_{n3} & \cdots & a_{nn} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n3} & \cdots & a_{nn} \end{vmatrix} \\ &+ a_{13} \begin{vmatrix} a_{21} & a_{22} & a_{24} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{34} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & a_{n4} & \cdots & a_{nn} \end{vmatrix} + \cdots + (-1)^{1+n} a_{1n} \begin{vmatrix} a_{21} & a_{23} & \cdots & a_{2,n-1} \\ a_{31} & a_{33} & \cdots & a_{3,n-1} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n3} & \cdots & a_{n,n-1} \end{vmatrix} \end{aligned}$$

Example 3.1:

1. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then $\det A = a_{11}(-1)^{1+1}A_{11} + a_{12}(-1)^{1+2}A_{12} = ad - bc$.

$$\begin{aligned} 2. \quad \det \begin{bmatrix} -3 & -2 & 4 \\ 4 & 3 & 1 \\ 0 & 2 & 4 \end{bmatrix} &= \begin{vmatrix} -3 & -2 & 4 \\ 4 & 3 & 1 \\ 0 & 2 & 4 \end{vmatrix} = (-3) \begin{vmatrix} 3 & 1 \\ 2 & 4 \end{vmatrix} - (-2) \begin{vmatrix} 4 & 1 \\ 0 & 4 \end{vmatrix} + 4 \begin{vmatrix} 4 & 3 \\ 0 & 2 \end{vmatrix} \\ &= -3(12 - 2) + 2(16 - 0) + 4(8 - 0) = 34, \end{aligned}$$

$$\text{where } M_{11} = \begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix} \text{ and } A_{11} = (-1)^{1+1} \det M_{11} = \begin{vmatrix} 3 & 1 \\ 2 & 4 \end{vmatrix},$$

$$M_{12} = \begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix} \text{ and } A_{12} = (-1)^{1+2} \det M_{12} = - \begin{vmatrix} 4 & 1 \\ 0 & 4 \end{vmatrix},$$

$$\text{and } M_{13} = \begin{bmatrix} 4 & 3 \\ 0 & 2 \end{bmatrix} \text{ and } A_{13} = (-1)^{1+3} \det M_{13} = \begin{vmatrix} 4 & 3 \\ 0 & 2 \end{vmatrix}.$$

3. Show that $\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = aei + bfg + cdh - afh - bdi - ceg.$

Solution:

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$$= a(ei - fh) - b(di - fg) + c(dh - eg) = aei + bfg + cdh - afh - bdi - ceg.$$

This formula can be easily remembered by using following diagram:

$$\begin{array}{ccccc} a & b & c & a & b \\ d & e & f & d & e \\ g & h & i & g & h \\ - & - & - & + & + \end{array}$$

However, this method cannot be generalized to higher order.

Example:

$$\det \begin{bmatrix} -3 & -2 & 4 \\ 4 & 3 & 1 \\ 0 & 2 & 4 \end{bmatrix} = (-3)(3)(4) + (-2)(1)(0) + 4(4)(2) - (4)(3)(0) - (-3)(1)(2) - (-2)(4)(4) = 34.$$

$$\begin{array}{ccccc} -3 & -2 & 4 & -3 & -2 \\ 4 & 3 & 1 & 4 & 3 \\ 0 & 2 & 4 & 0 & 2 \\ - & - & - & + & + \end{array}$$

4. $\det \begin{bmatrix} 2 & -1 & 0 & 0 \\ 3 & -3 & 2 & -2 \\ 4 & 2 & 0 & 0 \\ 2 & 0 & 0 & -1 \end{bmatrix} = 2 \begin{vmatrix} -3 & 2 & -2 \\ 2 & 0 & 0 \\ 0 & 0 & -1 \end{vmatrix} - (-1) \begin{vmatrix} 3 & 2 & -2 \\ 4 & 0 & 0 \\ 2 & 0 & -1 \end{vmatrix} + 0 - 0$

$$= 2 \left[-3 \begin{vmatrix} 0 & 0 \\ 0 & -1 \end{vmatrix} - 2 \begin{vmatrix} 2 & 0 \\ 0 & -1 \end{vmatrix} - 2 \begin{vmatrix} 2 & 0 \\ 0 & 0 \end{vmatrix} \right] + 3 \left[0 \begin{vmatrix} 0 & 0 \\ 0 & -1 \end{vmatrix} - 2 \begin{vmatrix} 4 & 0 \\ 2 & -1 \end{vmatrix} - 2 \begin{vmatrix} 4 & 0 \\ 2 & 0 \end{vmatrix} \right]$$

$$= 0 + -4(-2) - 0 + 0 + -2(-4) - 0 = 16.$$

Theorem 3.1: Cofactor Expansion Theorem

The determinant of an $n \times n$ matrix A can be computed by using the cofactor expansion along any row or any column of A . More precisely, if $A = [a_{ij}] \in M_n(\mathbb{R})$, then

the expansion along row i -th is $\det A = a_{i1}A_{i1} + a_{i2}A_{i2} + \dots + a_{in}A_{in}$.

The expansion along column j -th is $\det A = a_{1j}A_{1j} + a_{2j}A_{2j} + \dots + a_{nj}A_{nj}$.

For each $1 \leq i, j \leq n$, the number A_{ij} is called the **cofactor of a_{ij} in $\det A$** .

Example 3.2: Expand along the 3rd column:

$$\det \begin{bmatrix} 2 & -1 & 0 & 0 \\ 3 & -3 & 2 & -2 \\ 4 & 2 & 0 & 0 \\ 2 & 0 & 0 & -1 \end{bmatrix} = -2 \begin{vmatrix} 2 & -1 & 0 \\ 4 & 2 & 0 \\ 2 & 0 & -1 \end{vmatrix} = -2 \left[-1 \begin{vmatrix} 2 & -1 \\ 4 & 2 \end{vmatrix} \right] = 2[4 + 4] = 16.$$

Theorem 3.2:

Let $A = [a_{ij}] \in M_n(\mathbb{R})$. If A is either a **triangular or diagonal** matrix, then the determinant of A is equal to the product of the diagonal entries of A . That is, $\det A = a_{11}a_{22} \cdots a_{nn}$.

Example 3.3:

$$1. \det I_n = 1 \text{ and } \begin{vmatrix} 2 & 0 & 0 & 0 \\ 0 & -3 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & -1 \end{vmatrix} = (2)(-3)(5)(-1) = 30.$$

$$2. \begin{vmatrix} -1 & 0 & 0 \\ 2 & 4 & 0 \\ 1 & 3 & 2 \end{vmatrix} = (-1)(4)(2) = -8.$$

$$3. \begin{vmatrix} -2 & -1 & 2 & 0 \\ 0 & -3 & 3 & -2 \\ 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & -1 \end{vmatrix} = (-2)(-3)(0)(-1) = 0.$$

Theorem 3.3:

Let $A \in M_n(\mathbb{R})$. Then,

1. $\det A = \det A^T$.
2. If A contains a zero row (a row of zeros) or a zero column (a column of zeros), then $\det A = 0$.
3. If A contains two identical rows or two identical columns, then $\det A = 0$.

Example 3.4:

$$1. \begin{vmatrix} 0 & -1 & 2 & 0 \\ 2 & -3 & 3 & -2 \\ 0 & 0 & 4 & 1 \\ 0 & 0 & 2 & -1 \end{vmatrix} = \begin{vmatrix} 0 & 2 & 0 & 0 \\ -1 & -3 & 0 & 0 \\ 2 & 3 & 4 & 2 \\ 0 & -2 & 1 & -1 \end{vmatrix} = -2 \begin{vmatrix} -1 & 0 & 0 \\ 2 & 4 & 2 \\ 0 & 1 & -1 \end{vmatrix} = -2 \left[-1 \begin{vmatrix} 4 & 2 \\ 1 & -1 \end{vmatrix} \right] = 2[-4 - 2] = -12.$$

$$2. \begin{vmatrix} 1 & 3 & -1 \\ 2 & 4 & 1 \\ 1 & 3 & -1 \end{vmatrix} = 0 \text{ and } \begin{vmatrix} 2 & 0 & 3 & 0 \\ -1 & -3 & 1 & -3 \\ 2 & 4 & 2 & 4 \\ 0 & 5 & 3 & 5 \end{vmatrix} = 0.$$

Theorem 3.4:

Let $A, B \in M_n(\mathbb{R})$. Then,

1. If B is obtained from A by multiplying one row of A by a nonzero constant c , then $\det B = c \det A$.

In this case, $\det A = \frac{1}{c} \det B$. ($A \xrightarrow{R_i \rightarrow cR_i} B$)

2. If B is obtained from A by interchanging two rows, then $\det B = -\det A$. ($A \xrightarrow{R_i \leftrightarrow R_j} B$)

3. If B is obtained from A by adding a multiple of one row of A to another row, then $\det B = \det A$.

($A \xrightarrow{R_i \rightarrow R_i + cR_j} B$)

4. If $E \in M_n(\mathbb{R})$ is an elementary matrix, then $\det(EA) = \det(E) \det(A)$.

Proof of Part (3):

Let $A = [a_{ij}] \in M_n(\mathbb{R})$ and let B be a n -square matrix obtained from A by adding c times of the j -th row of A to the i -th row of A . Then

$$\det B = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{i1} + ca_{j1} & a_{i2} + ca_{j2} & \cdots & a_{in} + ca_{jn} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = (a_{i1} + ca_{j1})A_{i1} + (a_{i2} + ca_{j2})A_{i2} + \cdots + (a_{in} + ca_{jn})A_{in}$$

$$= (a_{i1}A_{i1} + a_{i2}A_{i2} + \cdots + a_{in}A_{in}) + c(a_{j1}A_{i1} + a_{j2}A_{i2} + \cdots + a_{jn}A_{in})$$

$$= \det(A) + c \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{i-1,1} & a_{i-1,2} & & a_{i-1,n} \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ a_{i+1,1} & a_{i+1,2} & & a_{i+1,n} \\ \vdots & \vdots & & \vdots \\ a_{j1} & a_{j2} & \cdots & a_{jn} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} = \det(A) + 0 = \det(A).$$

Remark 3.1:

By Theorem 3.4, we can use elementary row operations to transform a square matrix to triangular matrix and then by Theorem 3.2 to compute the determinant accordingly.

Example 3.5:

Use elementary row operations to find the determinant of the matrix $\begin{bmatrix} 3 & -1 & 1 & 1 \\ 3 & -1 & 2 & 1 \\ 0 & 2 & 4 & 3 \\ -3 & 1 & 1 & -2 \end{bmatrix}$.

Solution:

$$\begin{vmatrix} 3 & -1 & 1 & 1 \\ 3 & -1 & 2 & 1 \\ 0 & 2 & 4 & 3 \\ -3 & 1 & 1 & -2 \end{vmatrix} \xrightarrow{R_4 \rightarrow R_4 + R_1} \begin{vmatrix} 3 & -1 & 1 & 1 \\ 3 & -1 & 2 & 1 \\ 0 & 2 & 4 & 3 \\ 0 & 0 & 2 & -1 \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{vmatrix} 3 & -1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 4 & 3 \\ 0 & 0 & 2 & -1 \end{vmatrix}$$

$$\xrightarrow{R_2 \leftrightarrow R_3} - \begin{vmatrix} 3 & -1 & 1 & 1 \\ 0 & 2 & 4 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 \end{vmatrix} \xrightarrow{R_4 \rightarrow R_4 - 2R_3} - \begin{vmatrix} 3 & -1 & 1 & 1 \\ 0 & 2 & 4 & 3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{vmatrix} = -3(2)(1)(-1) = 6.$$

Example 3.6:

Let A and B be square matrices such that B can be obtained from A by elementary row operations:

$$A \xrightarrow{R_1 \rightarrow R_1 + \frac{2}{9}R_2} A_1 \xrightarrow{R_2 \leftrightarrow R_3} A_2 \xrightarrow{R_2 \rightarrow 4R_2} B.$$

Find $\det A$ if $B = \begin{bmatrix} 5 & 0 & 8 & -1 \\ 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 9 \\ 0 & 0 & 0 & 1/3 \end{bmatrix}$.

Solution:

Since B can be obtained from A by the given elementary row operations, then

$$\det B = 4 \det A_2 = 4(-\det A_1) = -4 \det A.$$

$$\text{Hence, } \det A = -\frac{1}{4} \det B = -\frac{1}{4}(5)(-2)(1)\left(-\frac{1}{3}\right) = \frac{5}{6}.$$

Remark 3.2:

Since $\det A = \det A^T$ for any n -square matrix A , then Theorem 3.4 holds if we change “row” to “column”. That is, let $A, B \in M_n(\mathbb{R})$. Then,

1. If B is obtained from A by multiplying one column of A by a nonzero constant k , then

$$\det B = k \det A. \quad (A \xrightarrow{C_i \rightarrow kC_i} B)$$

2. If B is obtained from A by interchanging two columns, then $\det B = -\det A$. $(A \xrightarrow{C_i \leftrightarrow C_j} B)$

3. If B is obtained from A by adding a multiple of one column of A to another column, then

$$\det B = \det A. \quad (A \xrightarrow{C_i \rightarrow C_i + kC_j} B)$$

4. If $E \in M_n(\mathbb{R})$ is an elementary matrix, then $\det AE = \det A \det E$.

Theorem 3.5:

Let $A \in M_n(\mathbb{R})$. Then A is invertible if and only if $\det A \neq 0$.

Example 3.7:

1. Let $A = \begin{bmatrix} 3 & -1 & 1 & 1 \\ 3 & -1 & 2 & 1 \\ 0 & 2 & 4 & 3 \\ -3 & 1 & 1 & -2 \end{bmatrix}$. From Example 3.5, $\det A = 6 \neq 0$. Hence, the matrix A is invertible.

2. Let $B = \begin{bmatrix} 1 & -3 & 1 \\ -2 & 6 & 1 \\ 1 & -3 & -1 \end{bmatrix}$. Then, the matrix B is not invertible since

$$\det B = \begin{vmatrix} 1 & -3 & 1 \\ -2 & 6 & 1 \\ 1 & -3 & -1 \end{vmatrix} \xrightarrow{C_2 \rightarrow -\frac{1}{3}C_2} \begin{vmatrix} 1 & 1 & 1 \\ -2 & -2 & 1 \\ 1 & 1 & -1 \end{vmatrix} = -3(0) = 0.$$

or

$$\begin{array}{ccccccc} 1 & -3 & 1 & 1 & -3 \\ -2 & 6 & 1 & -2 & 6 \\ 1 & -3 & -1 & 1 & -3 \\ - & - & - & + & + & + \end{array} \quad \text{Hence, } \det B = \begin{vmatrix} 1 & -3 & 1 \\ -2 & 6 & 1 \\ 1 & -3 & -1 \end{vmatrix} = -6 - 3 + 6 - 6 + 3 + 6 = 0.$$

Theorem 3.6:

Let $A, B \in M_n(\mathbb{R})$ and c be a scalar. Then,

1. $\det(cA) = c^n \det(A)$;
2. $\det(AB) = \det(A)\det(B)$; and
3. if A is invertible, then $\det(A^{-1}) = \frac{1}{\det(A)}$.

Example 3.8: Let $A = \begin{bmatrix} -2 & 0 & 4 \\ 4 & -6 & -8 \\ -4 & -2 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & 2 \end{bmatrix}$. Then

$$\det A = \det \left(\begin{bmatrix} 1 & 0 & -2 \\ -2 & 3 & 4 \\ 2 & 1 & -3 \end{bmatrix} \right) = (-2)^3 \det \begin{bmatrix} 1 & 0 & -2 \\ -2 & 3 & 4 \\ 2 & 1 & -3 \end{bmatrix} \xrightarrow{C_3 \rightarrow C_3 + 2C_1} -8 \det \begin{bmatrix} 1 & 0 & 0 \\ -2 & 3 & 0 \\ 2 & 1 & 1 \end{bmatrix} = -24$$

$$\text{and } \det B \xrightarrow{R_3 \rightarrow R_3 - R_1} \det \begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = (1)(-1)(1) = -1.$$

Hence, $\det AB = (\det A)(\det B) = 24 \neq 0$. Therefore, AB is invertible and $\det(AB)^{-1} = \frac{1}{\det AB} = \frac{1}{24}$.

3.2 Adjoint Matrix of A ($\text{adj} A$)

Definition 3.2: Adjoint of A

Let $A = [a_{ij}]$ be a n -square matrix. The transpose of the matrix of cofactors of the element of a_{ij} is called the **adjoint of A** , and is denoted by **$\text{adj} A$** .

That is, $\text{adj} A = [A_{ij}]^T$ where A_{ij} is the cofactor of a_{ij} in $\det A$.

$$\text{adj} A = \begin{bmatrix} A_{11} & A_{21} & \cdots & A_{n1} \\ A_{12} & A_{22} & \cdots & A_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ A_{1n} & A_{2n} & \cdots & A_{nn} \end{bmatrix}$$

Remark 3.3: Let A be an $n \times n$ matrix.

- (1) $\text{adj} A$ must exist.
- (2) $\text{adj} A$ is an $n \times n$ matrix.
- (3) $\text{adj} A$ is unique.
- (4) $\text{adj} A$ is the transpose of the matrix of cofactors of a_{ij} .

Example 3.9: Let $A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. Find $\text{adj} A$.

Solution. First calculate the minors,

$$\begin{aligned} M_{11} &= \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}, & M_{12} &= \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, & M_{13} &= \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}, \\ M_{21} &= \begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}, & M_{22} &= \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}, & M_{23} &= \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}, \\ M_{31} &= \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}, & M_{32} &= \begin{bmatrix} 3 & 2 \\ 1 & 1 \end{bmatrix}, & M_{33} &= \begin{bmatrix} 3 & 1 \\ 1 & 1 \end{bmatrix}, \end{aligned}$$

that is

$$\begin{aligned} |M_{11}| &= 1, & |M_{12}| &= 0, & |M_{13}| &= -1, \\ |M_{21}| &= -1, & |M_{22}| &= 1, & |M_{23}| &= 2, \\ |M_{31}| &= -3, & |M_{32}| &= 1, & |M_{33}| &= 5, \end{aligned}$$

then determine the cofactors,

$$\begin{aligned} A_{11} &= (-1)^{1+1}|M_{11}| = 1, & A_{12} &= (-1)^{1+2}|M_{12}| = 0, & A_{13} &= (-1)^{1+3}|M_{13}| = -1 \\ A_{21} &= (-1)^{2+1}|M_{21}| = 1, & A_{22} &= (-1)^{2+2}|M_{22}| = 1, & A_{23} &= (-1)^{2+3}|M_{23}| = -2 \\ A_{31} &= (-1)^{3+1}|M_{31}| = -3, & A_{32} &= (-1)^{3+2}|M_{32}| = -1, & A_{33} &= (-1)^{3+3}|M_{33}| = 5. \end{aligned}$$

Therefore the adjoint is

$$\text{adj} A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & -2 \\ -3 & -1 & 5 \end{bmatrix}^T = \begin{bmatrix} 1 & 1 & -3 \\ 0 & 1 & -1 \\ -1 & -2 & 5 \end{bmatrix}.$$

Properties of the adjoint

Theorem 3.7 *Let A be an $n \times n$ matrix. Then*

$$(1) \quad A(\operatorname{adj} A) = |A|I_n = (\operatorname{adj} A)A$$

$$(2) \quad |\operatorname{adj} A| = |A|^{n-1} \text{ if } |A| \neq 0$$

$$\text{Note that } |A|I_n = \begin{bmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{bmatrix}.$$

Theorem 3.8: *Let A and B be $n \times n$ matrices. Then*

$$(1) \quad \operatorname{adj}(A + B) \neq \operatorname{adj} A + \operatorname{adj} B$$

$$(2) \quad \operatorname{adj}(AB) = (\operatorname{adj} B)(\operatorname{adj} A)$$

$$(3) \quad \operatorname{adj}(A^{-1}) = (\operatorname{adj} A)^{-1} \text{ if } A \text{ is invertible}$$

$$(4) \quad \operatorname{adj}(A^\top) = (\operatorname{adj} A)^\top$$

$$(5) \quad \operatorname{adj}(\lambda A) = \lambda^{n-1}(\operatorname{adj} A) \text{ where } \lambda \in \mathbb{F}, \lambda \neq 0$$

Theorem 3.9: (A criterion for the inverse to exist and a formula to find the inverse).
Let A be an $n \times n$ matrix. Then A is invertible if and only if $\operatorname{adj} A$ is invertible.

If A is invertible then $A^{-1} = \frac{\operatorname{adj} A}{|A|}$.

Example 3.10: Let $A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. Is A invertible? If so, find A^{-1} by using the adjoint of A .

Solution. Since $|A| = 1 \neq 0$, A is invertible and

$$A^{-1} = \frac{\operatorname{adj} A}{|A|} = \frac{1}{1} \begin{bmatrix} 1 & 1 & -3 \\ 0 & 1 & -1 \\ -1 & -2 & 5 \end{bmatrix} = \begin{bmatrix} 1 & 1 & -3 \\ 0 & 1 & -1 \\ -1 & -2 & 5 \end{bmatrix}.$$

3.3 Cramer's Rule

Theorem 3.10: Cramer's Rule

Let A be an $n \times n$ invertible matrix. Then the system of linear equations $AX = B$ where

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix},$$

has a unique solution and the unique solution of the system is given by $x_i = \frac{\det A_i}{\det A}$, where, for each

$1 \leq i \leq n$, let A_i be the matrix obtained from A by replacing the i -th column of A by the column matrix

$$B. \text{ That is, } \det(A_i) = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1,i-1} & b_1 & a_{1,i+1} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2,i-1} & b_2 & a_{2,i+1} & \cdots & a_{2n} \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{n,i-1} & b_n & a_{n,i+1} & \cdots & a_{nn} \end{vmatrix}.$$

Example 3.11:

The system of linear equations

$$\begin{aligned} x + y + z &= 6 \\ x + 2y + 3z &= 14 \\ x + 4y + 9z &= 36 \end{aligned}$$

is equivalent to $AX = B$ that is $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 14 \\ 36 \end{bmatrix}$.

Since $\det A = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{vmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 0 & 3 & 8 \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 3 & 8 \end{vmatrix} = 1(8 - 6) = 2 \neq 0$, then the system has a

unique solution.

$$\det A_1 = \begin{vmatrix} 6 & 1 & 1 \\ 14 & 2 & 3 \\ 36 & 4 & 9 \end{vmatrix} \xrightarrow{R_3 \rightarrow R_3 - 4R_1} \begin{vmatrix} 6 & 1 & 1 \\ 14 & 2 & 3 \\ 12 & 0 & 5 \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{vmatrix} 6 & 1 & 1 \\ 2 & 0 & 1 \\ 12 & 0 & 5 \end{vmatrix} = -1(10 - 12) = 2.$$

$$\det A_2 = \begin{vmatrix} 1 & 6 & 1 \\ 1 & 14 & 3 \\ 1 & 36 & 9 \end{vmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{vmatrix} 1 & 6 & 1 \\ 1 & 14 & 3 \\ 0 & 30 & 8 \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{vmatrix} 1 & 6 & 1 \\ 0 & 8 & 2 \\ 0 & 30 & 8 \end{vmatrix} = 1(64 - 60) = 4.$$

$$\det A_3 = \begin{vmatrix} 1 & 1 & 6 \\ 1 & 2 & 14 \\ 1 & 4 & 36 \end{vmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{vmatrix} 1 & 1 & 6 \\ 1 & 2 & 14 \\ 0 & 3 & 30 \end{vmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{vmatrix} 1 & 1 & 6 \\ 0 & 1 & 8 \\ 0 & 3 & 30 \end{vmatrix} = 1(30 - 24) = 6.$$

By using Cramer's rule, the solution is given by

$$x = \frac{\det A_1}{\det A} = \frac{2}{2} = 1, \quad y = \frac{\det A_2}{\det A} = \frac{4}{2} = 2 \quad \text{and} \quad z = \frac{\det A_3}{\det A} = \frac{6}{2} = 3.$$

3.4 Linearly independent row vectors and rank

Definition 3.3:

Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be n vectors in a vector set V (that is $V = \{ \mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n \}$).

- (i) We said that the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are **linearly independent** if

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n = \mathbf{0} \text{ implies } \lambda_1 = \lambda_2 = \dots = \lambda_n = 0.$$
- (ii) $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are **linearly dependent** if there exist numbers $\lambda_1, \lambda_2, \dots, \lambda_n$, not all of them zero, such that $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n = \mathbf{0}$.

Example 3.12:

- (1) Let $V = \{(1, 0, 0), (1, 2, 0), (1, 0, 1)\}$. Then,

$$\begin{aligned} \lambda_1(1, 0, 0) + \lambda_2(1, 2, 0) + \lambda_3(1, 0, 1) &= \mathbf{0} \Rightarrow (\lambda_1 + \lambda_2 + \lambda_3, 2\lambda_2, \lambda_3) = (0, 0, 0) \\ \Rightarrow \lambda_1 + \lambda_2 + \lambda_3 &= 0 \\ 2\lambda_2 &= 0 \\ \lambda_3 &= 0 \\ \Rightarrow \lambda_2 = 0, \lambda_3 = 0 \text{ and } \lambda_1 &= 0. \end{aligned}$$

Therefore, $(1, 0, 0), (1, 2, 0), (1, 0, 1)$ are linearly independent.

- (2) Let $V = \{(1, 0, 0), (1, 2, 2), (1, -4, -4)\}$.

Since $3(1, 0, 0) - 2(1, 2, 2) - (1, -4, -4) = \mathbf{0}$, then, there exists $\lambda_i \neq 0, i = 1, 2, 3$ such that $\lambda_1(1, 0, 0) + \lambda_2(1, 2, 2) + \lambda_3(1, -4, -4) = \mathbf{0}$,

Therefore, $(1, 0, 0), (1, 2, 2), (1, -4, -4)$ are linearly dependent.

Definition 3.4:

Let $A = [a_{ij}] \in M_{m,n}(\mathbb{R})$ and let $\mathbf{v}$ be a row vector of A .

Then, the maximum number of independent row vectors of A is called the **rank of A** and is denoted by $\rho(A)$ or **rank A** .

Example 3.13:

Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 2 & 2 \\ 1 & -4 & -4 \end{bmatrix}$ where $\mathbf{v}_1 = (1, 0, 0), \mathbf{v}_2 = (1, 2, 2)$ and $\mathbf{v}_3 = (1, -4, -4)$.

From Example 3.10, $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ are linearly dependent.

However, $\mathbf{v}_1$ and $\mathbf{v}_2$ (and also $\mathbf{v}_1$ and $\mathbf{v}_3$) are linearly independent.

$$\begin{aligned} \lambda_1(1, 0, 0) + \lambda_2(1, 2, 2) &= \mathbf{0} \Rightarrow (\lambda_1 + \lambda_2, 2\lambda_2, 2\lambda_2) = (0, 0, 0) \\ \Rightarrow \lambda_1 + \lambda_2 &= 0 \text{ and } 2\lambda_2 = 0 \\ \Rightarrow \lambda_2 &= 0 \text{ and } \lambda_1 = 0 \\ \Rightarrow \mathbf{v}_1 \text{ and } \mathbf{v}_2 &\text{ are linearly independent.} \end{aligned}$$

Similar for $\mathbf{v}_1$ and $\mathbf{v}_3$, $\lambda_1(1, 0, 0) + \lambda_2(1, -4, -4) = \mathbf{0} \Rightarrow (\lambda_1 + \lambda_2, -4\lambda_2, -4\lambda_2) = (0, 0, 0)$

$$\begin{aligned} \Rightarrow \lambda_1 + \lambda_2 &= 0 \text{ and } -4\lambda_2 = 0 \\ \Rightarrow \lambda_2 &= 0 \text{ and } \lambda_1 = 0 \\ \Rightarrow \mathbf{v}_1 \text{ and } \mathbf{v}_3 &\text{ are linearly independent.} \end{aligned}$$

The maximum number of independent row vectors = 2.

Therefore, the rank of $A = \rho(A) = 2$.

Remark 3.3:

Let $V = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. If there exists a zero row vector in V (that is $\mathbf{v}_i = \mathbf{0}$), then $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly dependent. This is because if $\mathbf{v}_i = \mathbf{0}$, then λ_i does not have to be zero so that $0\mathbf{v}_1 + 0\mathbf{v}_2 + \dots + \lambda_i \mathbf{v}_i + 0\mathbf{v}_{i+1} + \dots + 0\mathbf{v}_n = \mathbf{0}$.

Theorem 3.11: If $A = [a_{ij}] \in M_{m,n}(\mathbb{R})$ is a row echelon matrix, then

the rank of $A = \rho(A)$ = the number of non-zero row(s) in A .

Example 3.14:

Let A be a matrix formed by the row vectors set V .

(a) $V_1 = \{(0, 1, 0), (1, 0, 2), (2, 0, -1)\}$.

(b) $V_2 = \{(1, -1, 4, 2), (0, 1, 3, 2), (3, -2, 15, 8)\}$.

- (i) Determine the rank of A .
- (ii) Determine whether the row vectors set V is linearly dependent or is linearly independent.
- (iii) The maximum number of linearly independent row vectors in V .

Solution:

$$(a) \quad A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 2 \\ 2 & 0 & -1 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 2 & 0 & -1 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - 2R_1} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & -5 \end{bmatrix} \text{ (R.E.M)}$$

- (i) The number of non-zero rows in the row echelon matrix = 3.
 $\therefore$ The rank of $A = 3$.
- (ii) There is no zero row in the row echelon matrix, the row vectors set V_1 is linearly independent.
- (iii) The maximum number of linearly independent row vectors in $V_1 = 3$.

$$(a) \quad A = \begin{bmatrix} 1 & -1 & 4 & 2 \\ 0 & 1 & 3 & 2 \\ 3 & -2 & 15 & 8 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - 3R_1} \begin{bmatrix} 1 & -1 & 4 & 2 \\ 0 & 1 & 3 & 2 \\ 0 & 1 & 3 & 2 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{bmatrix} 1 & -1 & 4 & 2 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{ (R.E.M)}$$

- (i) The number of non-zero rows in the row echelon matrix = 2.
 $\therefore$ The rank of $A = 2$.
- (ii) There is no zero row in the row echelon matrix, the row vectors set V_2 is linearly dependent.
- (iii) The maximum number of linearly independent row vectors in $V_2 = 2$.

Chapter 4 Diagonalization

4.1 Eigenvalues and Eigenvectors

Definition 4.1:

Let $A \in M_n(\mathcal{R})$. A scalar λ is called an **eigenvalue of A** if there exists a nonzero column vector $X \in M_{n,1}(\mathcal{R})$ such that $AX = \lambda X$. X is called an **eigenvector of A** corresponding to (or associated with) the eigenvalue λ .

Example 4.1:

$$\text{Let } A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}, X_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \text{ and } X_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

$$AX_1 = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = -1 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = -X_1.$$

$\therefore X_1$ is an eigenvector of A corresponds to the eigenvalue $\lambda_1 = -1$.

$$AX_2 = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 5X_2.$$

$\therefore X_2$ is an eigenvector of A corresponds to the eigenvalue $\lambda_2 = 5$.

Example 4.2: Let $A \in M_n(\mathcal{R})$. Show that

- (i) If $\lambda \in \mathcal{R}$ is an eigenvalue of A , then λ^n is an eigenvalue of A^n for every positive integer n .
- (ii) If $X \in M_{n,1}(\mathcal{R})$ is an eigenvector of A corresponding to the eigenvalue $\lambda \in \mathcal{R}$, then, for every nonzero scalar $\alpha \in \mathcal{R}$, the vector αX is an eigenvector of A corresponds to the eigenvalue λ .

Solution:

- (i) λ is an eigenvalue of A if there exists a nonzero column vector such that $AX = \lambda X$.
 $A^n X = A^{n-1}(AX) = A^{n-1}(\lambda X) = \lambda A^{n-2}(AX) = \lambda A^{n-2}(\lambda X) = \lambda^2 A^{n-2} X = \lambda^2 A^{n-3}(AX) = \lambda^2 A^{n-3}(\lambda X)$
 $= \lambda^3 A^{n-3} X = \dots = \lambda^{n-1}(AX) = \lambda^{n-1}(\lambda X) = \lambda^n X.$

Hence, λ^n is an eigenvalue of A^n for every positive integer n .

- (ii) For every nonzero scalar $\alpha \in \mathcal{R}$, $AX = \lambda X \Rightarrow \alpha(AX) = \alpha(\lambda X) \Rightarrow A(\alpha X) = \lambda(\alpha X).$

Hence, the vector αX is an eigenvector of A corresponds to the eigenvalue λ .

Theorem 4.2:

Let $A \in M_n(\mathcal{R})$. A scalar λ is an eigenvalue of A if and only if $\det(A - \lambda I_n) = 0$.

Proof:

- λ is an eigenvalue of $A \Rightarrow$ there exists a nonzero column vector $X \in M_{n,1}$ such that $AX = \lambda X$
 $\Rightarrow$ there exists a nonzero column vector $X \in M_{n,1}$ such that $(A - \lambda I_n)X = 0$
 $\Rightarrow$ the homogenous system has a nontrivial solution ($\because X \neq 0$)
 $\Rightarrow$ the matrix $(A - \lambda I_n)$ is singular (not invertible) $\Rightarrow \det(A - \lambda I_n) = 0$.

Definition 4.3:

Let $A \in M_n(\mathcal{R})$. Then, $p(\lambda) = \det(A - \lambda I_n)$ is called a **characteristic polynomial of A** and $p(\lambda) = \det(A - \lambda I_n) = 0$ is the **characteristic equation of A** .

Procedure to determine eigenvalues and eigenvector of A

1. For a given $A \in M_n$, determine the characteristic polynomial

$$P(\lambda) = \det(A - \lambda I_n) = \begin{vmatrix} a_{11} - \lambda & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} - \lambda & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} - \lambda \end{vmatrix}.$$

2. Solve the characteristic equation $p(\lambda) = 0$ and determine all the eigenvalues λ_i .
3. For each λ_i , solve for e_i in equation $Ae = \lambda_i e_i$, or equivalently, $(A - \lambda_i I_n)e_i = 0$, and find an eigenvector e_i corresponding to λ_i .

Example 5.3:

Determine the characteristic equation of $A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$.

Then, for each λ_i , find an eigenvector e_i corresponding to λ_i .

Solution:

$$A - \lambda I_3 = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1-\lambda & 1 & -2 \\ -1 & 2-\lambda & 1 \\ 0 & 1 & -1-\lambda \end{bmatrix}$$

So, the characteristic polynomial is

$$\begin{aligned} p(\lambda) &= \det(A - \lambda I_3) \\ &= \begin{vmatrix} 1-\lambda & 1 & -2 \\ -1 & 2-\lambda & 1 \\ 0 & 1 & -1-\lambda \end{vmatrix} \\ &= (1-\lambda) \begin{vmatrix} 2-\lambda & 1 \\ 1 & -1-\lambda \end{vmatrix} - (-1) \begin{vmatrix} 1 & -2 \\ 1 & -1-\lambda \end{vmatrix} \quad (\text{Expanding along column 1}) \\ &= (1-\lambda)[(2-\lambda)(-1-\lambda) - 1] + [(-1-\lambda) + 2] \\ &= \lambda^3 - 2\lambda^2 - \lambda + 2. \end{aligned}$$

Let $p(\lambda) = \det(A - \lambda I_n) = 0$.

Then $\lambda^3 - 2\lambda^2 - \lambda + 2 = (\lambda - 2)(\lambda - 1)(\lambda + 1) = 0$.

$\therefore \lambda_1 = 2, \lambda_2 = 1, \lambda_3 = -1$.

Next, find eigenvectors e_i corresponding to λ_i by solving the system $(A - \lambda_i I_3)e_i = 0$.

That is, Solve $\begin{bmatrix} 1-\lambda & 1 & -2 \\ -1 & 2-\lambda & 1 \\ 0 & 1 & -1-\lambda \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$

Case (i) $\lambda_1 = 2$.

Augmented matrix:

$$\begin{bmatrix} 1-2 & 1 & -2 & 0 \\ -1 & 2-2 & 1 & 0 \\ 0 & 1 & -1-2 & 0 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{bmatrix} -1 & 1 & -2 & 0 \\ 0 & -1 & 3 & 0 \\ 0 & 1 & -3 & 0 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + R_2} \begin{bmatrix} -1 & 0 & 1 & 0 \\ 0 & -1 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\therefore 0 = 0$$

$$-x_2 + 3x_3 = 0 \Rightarrow x_2 = 3x_3 \text{ and } x_3 = x_1$$

$$-x_1 + x_3 = 0$$

Let $x_1 = k$. Then $x_2 = 3k$, $x_3 = k$, $k \neq 0$.

$\therefore$ The set of eigenvectors corresponding to $\lambda_1 = 2$ is given by $\left\{ \begin{pmatrix} k \\ 3-k \\ -3-k \end{pmatrix}, k \neq 0 \right\}$.

Case (ii) $\lambda_2 = 1$.

Augmented matrix:

$$\begin{bmatrix} 1-1 & 1 & -2 & 0 \\ -1 & 2-1 & 1 & 0 \\ 0 & 1 & -1-1 & 0 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_1} \begin{bmatrix} 0 & 1 & -2 & 0 \\ -1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 + R_1} \begin{bmatrix} 0 & 1 & -2 & 0 \\ -1 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\therefore 0 = 0$$

$$-x_1 + 3x_3 = 0 \Rightarrow x_1 = 3x_3 \text{ and } x_2 = 2x_3$$

$$x_2 - 2x_3 = 0$$

Let $x_3 = k$. Then $x_1 = 3k$, $x_2 = 2k$, $k \neq 0$.

$\therefore$ The set of eigenvectors corresponding to $\lambda_2 = 1$ is given by $\left\{ \begin{pmatrix} 3k \\ 2k \\ k \end{pmatrix}, k \neq 0 \right\}$.

Case (iii) $\lambda_3 = -1$.

Augmented matrix:

$$\begin{bmatrix} 1+1 & 1 & -2 & 0 \\ -1 & 2+1 & 1 & 0 \\ 0 & 1 & -1+1 & 0 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 + \frac{1}{2}R_1} \begin{bmatrix} 2 & 1 & -2 & 0 \\ 0 & \frac{7}{2} & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - \frac{7}{2}R_3} \begin{bmatrix} 2 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

$$\therefore x_2 = 0$$

$$0 = 0 \Rightarrow x_2 = 0 \text{ and } x_1 = x_3$$

$$2x_1 - 2x_3 = 0$$

Let $x_1 = k$. Then $x_3 = k$, $k \neq 0$.

$\therefore$ The set of eigenvectors corresponding to $\lambda_2 = -1$ is given by $\left\{ \begin{pmatrix} k \\ 0 \\ k \end{pmatrix}, k \neq 0 \right\}$.

Example 4.4:

Determine all the eigenvalues and eigenvectors of $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$.

Solution:

The characteristic polynomial is

$$\begin{aligned} p(\lambda) &= \det(A - \lambda I_2) \\ &= \begin{vmatrix} \cos \theta - \lambda & -\sin \theta \\ \sin \theta & \cos \theta - \lambda \end{vmatrix} \\ &= (\cos \theta - \lambda)^2 + \sin^2 \theta \\ &= \lambda^2 - 2\lambda \cos \theta + 1. \end{aligned}$$

Let $p(\lambda) = \det(A - \lambda I_n) = 0$.

So, $\lambda^2 - 2\lambda \cos \theta + 1 = 0$

$$\therefore \lambda = \frac{2\cos \theta \pm \sqrt{4\cos^2 \theta - 4}}{2} = \cos \theta \pm i \sin \theta.$$

Next, find eigenvectors e_i corresponding to λ_i by solving the system $(A - \lambda_i I_2)e_i = 0$.

That is, solve $\begin{bmatrix} \cos \theta - \lambda & -\sin \theta \\ \sin \theta & \cos \theta - \lambda \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$.

Case (i) $\lambda_1 = \cos \theta + i \sin \theta$.

Augmented matrix:

$$\begin{bmatrix} -i \sin \theta & -\sin \theta & 0 \\ \sin \theta & -i \sin \theta & 0 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - iR_1} \begin{bmatrix} -i \sin \theta & -\sin \theta & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore 0 = 0$$

$$(-i \sin \theta)x_1 + (-\sin \theta)x_2 = 0 \Rightarrow x_2 = -ix_1.$$

Let $x_1 = k$. Then $x_2 = -ik$, $k \neq 0$.

$\therefore$ The set of eigenvectors corresponding to $\lambda_1 = \cos \theta + i \sin \theta$ is given by $\left\{ \begin{pmatrix} k \\ -ik \end{pmatrix}, k \neq 0 \right\}$.

Case (ii) $\lambda_2 = \cos \theta - i \sin \theta$.

Augmented matrix:

$$\begin{bmatrix} i \sin \theta & -\sin \theta & 0 \\ \sin \theta & i \sin \theta & 0 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 + iR_1} \begin{bmatrix} i \sin \theta & -\sin \theta & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore 0 = 0$$

$$(i \sin \theta)x_1 - (\sin \theta)x_2 = 0 \Rightarrow x_2 = ix_1.$$

Let $x_1 = k$. Then $x_2 = ik$, $k \neq 0$.

$\therefore$ The set of eigenvectors corresponding to $\lambda_2 = \cos \theta - i \sin \theta$ is given by $\left\{ \begin{pmatrix} k \\ ik \end{pmatrix}, k \neq 0 \right\}$.

4.2 Diagonalization

Recall that a matrix $D = (d_{ij}) \in M_n$ is called a **diagonal matrix** if $d_{ij} = 0$ for all $i \neq j$. That is,

$$D = \begin{bmatrix} d_{11} & 0 & \cdots & 0 \\ 0 & d_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & d_{nn} \end{bmatrix}.$$

Properties of diagonal matrix:

Let D and W are two diagonal matrices of order n .

$$\text{That is, } D = \begin{bmatrix} d_{11} & 0 & \cdots & 0 \\ 0 & d_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & d_{nn} \end{bmatrix} \text{ and } W = \begin{bmatrix} w_{11} & 0 & \cdots & 0 \\ 0 & w_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & w_{nn} \end{bmatrix}.$$

$$1. \quad DW \text{ and } WD \text{ are also diagonal matrix and } DW = WD = \begin{bmatrix} w_{11}d_{11} & 0 & \cdots & 0 \\ 0 & w_{22}d_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & w_{nn}d_{nn} \end{bmatrix}.$$

2. The determinant of $D = \det(D) = d_{11}d_{22}d_{33}\dots d_{nn}$.

3. If all of the diagonal elements of D are nonzero (that is $d_{ii} \neq 0 \forall i = 1, 2, \dots, n$), then D is

$$\text{invertible (or nonsingular) and } D^{-1} = \begin{bmatrix} \frac{1}{d_{11}} & 0 & \cdots & 0 \\ 0 & \frac{1}{d_{22}} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \frac{1}{d_{nn}} \end{bmatrix}.$$

4. The eigenvalues of D are its diagonal elements, that is $d_{11}, \dots, d_{nn}$.

Of course, most square matrices are not diagonal, but some are related to diagonal matrices in a way we will find useful in solving problems.

Definition 4.5:

Let $A, B \in M_n(\mathcal{R})$. A matrix A is similar to B , denoted $A \sim B$, if there exists an invertible matrix $P \in M_n(\mathcal{R})$ such that $A = P^{-1}BP$.

Definition 4.6:

Let $A \in M_n(\mathcal{R})$. Then, A is **diagonalizable** $\Leftrightarrow$ there exists an n -square matrix P such that $P^{-1}AP$ is a diagonal matrix. (That is, $P^{-1}AP = D$ where D is a diagonal matrix.)

When such a matrix P exists, we say that P **diagonalizes** A .

Example 4.5:

Let $A = \begin{bmatrix} 3 & 2 \\ 3 & -2 \end{bmatrix}$. There exists a matrix $P = \begin{bmatrix} 2 & 1 \\ 1 & -3 \end{bmatrix}$ such that

$$P^{-1}AP = \frac{1}{7} \begin{bmatrix} 3 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 3 & -2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & -3 \end{bmatrix} = \frac{1}{7} \begin{bmatrix} 3 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 8 & -3 \\ 4 & 9 \end{bmatrix} = \begin{bmatrix} 4 & 0 \\ 0 & -3 \end{bmatrix} = \text{diagonal matrix.}$$

Therefore, P diagonalizes A .

Theorem 4.7:

Let $A \in M_n(\mathcal{R})$. Then, A is diagonalizable $\Leftrightarrow A$ has n linearly independent eigenvectors.

Theorem 4.8:

n column vector $v_1, v_2, \dots, v_n$ are linearly independent

$\Leftrightarrow$ The matrix $A = (v_1, v_2, \dots, v_n)$ is invertible (nonsingular) $\Leftrightarrow \det(A) \neq 0$.

Procedure to determine a matrix P that diagonalizes the matrix A :

1. Determine the characteristic equation of A , that is $p(\lambda) = \det(A - \lambda I_n) = 0$.
2. Solve the characteristic equation $p(\lambda) = 0$ and determine all the eigenvalues λ_i .
3. For each λ_i , solve $(A - \lambda_i I_n)e_i = 0$ to find an eigenvector e_i corresponding to λ_i .
4. Form the matrix $P = [e_1 \mid e_2 \mid \dots \mid e_n]$.
5. Determine whether $|P|$ is nonzero.

$$6. \quad \text{If } |P| \neq 0, \text{ then } P^{-1}AP = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \lambda_n \end{bmatrix}.$$

Example 4.6:

Let $A = \begin{bmatrix} -1 & 4 \\ 0 & 3 \end{bmatrix}$. Find a matrix P that diagonalizes A and find the diagonal matrix D such that $D = P^{-1}AP$.

Solution:

The characteristic equation is $p(\lambda) = \det(A - \lambda I_2) = 0$.

$$\det(A - \lambda I_2) = \begin{vmatrix} -1-\lambda & 4 \\ 0 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^2 - 2\lambda - 3 = 0$$

$$\Rightarrow (\lambda - 3)(\lambda + 1) = 0 \Rightarrow \lambda = -1, 3.$$

Next, find eigenvectors e_i corresponding to λ_i by solving the system $(A - \lambda_i I_2)e_i = 0$.

That is, solve $\begin{bmatrix} -1-\lambda & 4 \\ 0 & 3-\lambda \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$

Case (i) $\lambda_1 = -1$.

Augmented matrix:

$$\begin{bmatrix} 0 & 4 & 0 \\ 0 & 4 & 0 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{bmatrix} 0 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore 4x_2 = 0 \Rightarrow x_2 = 0.$$

$\therefore$ The set of eigenvectors corresponding to $\lambda_1 = -1$ is given by $\left\{ \begin{pmatrix} k \\ 0 \end{pmatrix}, k \neq 0 \right\}$.

$\therefore$ When $k = 1$, $e_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is an eigenvector corresponding to $\lambda_1 = -1$.

Case (ii) $\lambda_2 = 3$.

$$\text{Augmented matrix is } \begin{bmatrix} -4 & 4 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$\therefore -4x_1 + 4x_2 = 0 \Rightarrow x_1 = x_2.$$

$\therefore$ The set of eigenvectors corresponding to $\lambda_2 = 3$ is given by $\left\{ \begin{pmatrix} k \\ k \end{pmatrix}, k \neq 0 \right\}$.

$\therefore$ When $k = 1$, $e_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigenvector corresponding to $\lambda_2 = 3$.

$$\text{Let } P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}. \text{ Since } |P| = \begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1 \neq 0, \text{ then } P^{-1} \text{ exists and } P^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}.$$

$$\text{Since } P^{-1}AP = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 4 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix} = D.$$

$$\text{Therefore } P = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \text{ diagonalizes the matrix } A \text{ and } D = P^{-1}AP = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}.$$

Example 4.7:

$$\text{Let } A = \begin{bmatrix} -1 & 0 & 5 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}. \text{ Find a matrix } P \text{ that diagonalizes } A \text{ and find the diagonal matrix } D \text{ such that}$$

$$D = P^{-1}AP.$$

Solution:

The characteristic polynomial of A is $p(\lambda) = \det(A - \lambda I_3)$, that is

$$p(\lambda) = \begin{vmatrix} -1-\lambda & 0 & 5 \\ 0 & 1-\lambda & 0 \\ 0 & 0 & -2-\lambda \end{vmatrix} = (-1-\lambda)(1-\lambda)(-2-\lambda),$$

$$\text{and } p(\lambda) = 0 \Rightarrow \lambda = -1, 1, -2.$$

Next, find eigenvectors e_i corresponding to λ_i by solving the system $(A - \lambda_i I_3)e_i = 0$.

$$\text{That is, Solve } \begin{bmatrix} -1-\lambda & 0 & 5 \\ 0 & 1-\lambda & 0 \\ 0 & 0 & -2-\lambda \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Case (i) $\lambda_1 = -1$.

Augmented matrix:

$$\begin{bmatrix} -1+1 & 0 & 5 & 0 \\ 0 & 1+1 & 0 & 0 \\ 0 & 0 & -2+1 & 0 \end{bmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + R_2 / 5} \begin{bmatrix} 0 & 2 & 0 & 0 \\ 0 & 0 & 5 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\therefore x_2 = 0 \text{ and } x_3 = 0.$$

Let $x_1 = k$ be any nonzero number.

$\therefore$ The set of eigenvectors corresponding to $\lambda_1 = -1$ is given by $\left\{ \begin{pmatrix} k \\ 0 \\ 0 \end{pmatrix}, k \neq 0 \right\}$.

When $k = 1$, $e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ is an eigenvector corresponding to $\lambda_1 = -1$.

Case (ii) $\lambda_2 = 1$.

Augmented matrix:

$$\begin{bmatrix} -1-1 & 0 & 5 & 0 \\ 0 & 1-1 & 0 & 0 \\ 0 & 0 & -2-1 & 0 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} -2 & 0 & 5 & 0 \\ 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

$$\therefore -2x_1 + 5x_3 = 0 \text{ and } -3x_3 = 0 \Rightarrow x_3 = 0 \text{ and } x_1 = 0.$$

Let $x_2 = k$ be any nonzero number.

$\therefore$ The set of eigenvectors corresponding to $\lambda_2 = 1$ is given by $\left\{ \begin{pmatrix} 0 \\ k \\ 0 \end{pmatrix}, k \neq 0 \right\}$.

When $k = 1$, $e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ is an eigenvector corresponding to $\lambda_2 = 1$.

Case (iii) $\lambda_3 = -2$.

Augmented matrix:

$$\begin{bmatrix} -1+2 & 0 & 5 & 0 \\ 0 & 1+2 & 0 & 0 \\ 0 & 0 & -2+2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 5 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

$$\therefore x_1 + 5x_3 = 0 \text{ and } 3x_2 = 0 \Rightarrow x_1 = -5x_3 \text{ and } x_2 = 0.$$

Let $x_3 = k$ be any nonzero number.

$\therefore$ The set of eigenvectors corresponding to $\lambda_3 = -2$ is given by $\left\{ \begin{pmatrix} -5k \\ 0 \\ k \end{pmatrix}, k \neq 0 \right\}$.

When $k = 1$, $e_3 = \begin{pmatrix} -5 \\ 0 \\ 1 \end{pmatrix}$ is an eigenvector corresponding to $\lambda_3 = -2$.

Let $P = \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$. Since $|P| = 1 \neq 0$, then P^{-1} exists.

Therefore P diagonalizes the matrix A and $D = P^{-1}AP = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{bmatrix}$.

Remark 4.1:

If a matrix A has n distinct eigenvalues, then A is diagonalizable.

However, there exist many matrices which are diagonalizable even though they have some repeated eigenvalues. In these matrices, all we need are n linearly independent eigenvectors of a matrix (that is, determine a suitable P where $|P| \neq 0$).

Example 4.8:

Let $A = \begin{bmatrix} 5 & -4 & 4 \\ 12 & -11 & 12 \\ 4 & -4 & 5 \end{bmatrix}$. Find a matrix P that diagonalizes A and find the diagonal matrix D such that $D = P^{-1}AP$.

Solution:

The characteristic polynomial of A is

$$p(\lambda) = \det(A - \lambda I_3) = \begin{vmatrix} 5-\lambda & -4 & 4 \\ 12 & -11-\lambda & 12 \\ 4 & -4 & 5-\lambda \end{vmatrix} = \begin{vmatrix} 1-\lambda & 0 & -1+\lambda \\ 12 & -11-\lambda & 12 \\ 4 & -4 & 5-\lambda \end{vmatrix} = \begin{vmatrix} 0 & 0 & -1+\lambda \\ 24 & -11-\lambda & 12 \\ 9-\lambda & -4 & 5-\lambda \end{vmatrix} \\ = \dots = (-1+\lambda)(1-\lambda)(3+\lambda).$$

$$p(\lambda) = 0 \Rightarrow \lambda = 1, 1, -3.$$

Next, find eigenvectors e_i corresponding to λ_i by solving the system $(A - \lambda_i I_3)e_i = 0$.

Case (i) $\lambda_1 = \lambda_2 = 1$.

Augmented matrix:

$$\left[\begin{array}{cccc} 5-1 & -4 & 4 & 0 \\ 12 & -11-1 & 12 & 0 \\ 4 & -4 & 5-1 & 0 \end{array} \right] \xrightarrow{R_2 \rightarrow R_2 - 3R_1} \left[\begin{array}{cccc} 4 & -4 & 4 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

$$\therefore x_1 - x_2 + x_3 = 0 \Rightarrow x_1 = x_2 - x_3.$$

Let $x_2 = s, x_3 = t$ be any numbers. Then $x_1 = s - t$.

$\therefore$ The set of eigenvectors corresponding to $\lambda_1 = 1$ is given by $\left\{ \begin{pmatrix} s-t \\ s \\ t \end{pmatrix}, (s, t) \neq (0, 0) \right\}$.

When $s = 1$ and $t = 0$ and, $e_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$ is an eigenvector corresponding to $\lambda_1 = 1$.

When $s = 0$ and $t = 1$ and, $e_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$ is also an eigenvector corresponding to $\lambda_1 = 1$.

(Note that e_1 and e_2 are linearly independent eigenvectors.)

Case (ii) $\lambda_3 = -3$.

Augmented matrix:

$$\begin{bmatrix} 5+3 & -4 & 4 & 0 \\ 12 & -11+3 & 12 & 0 \\ 4 & -4 & 5+3 & 0 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - 3R_1/2} \begin{bmatrix} 8 & -4 & 4 & 0 \\ 0 & -2 & 6 & 0 \\ 0 & -2 & 6 & 0 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - R_2} \begin{bmatrix} 2 & -1 & 1 & 0 \\ 0 & -2 & 6 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\therefore 2x_1 - x_2 + x_3 = 0$$

$$-x_2 + 3x_3 = 0 \Rightarrow x_2 = 3x_3 \text{ and } x_1 = (x_2 - x_3)/2 = x_3.$$

Let $x_3 = k$ be any nonzero number.

$\therefore$ The set of eigenvectors corresponding to $\lambda_3 = -3$ is given by $\left\{ \begin{pmatrix} k \\ 3k \\ k \end{pmatrix}, k \neq 0 \right\}$.

When $k = 1$, $e_3 = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}$ is an eigenvector corresponding to $\lambda_3 = -3$.

Let $P = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 3 \\ 0 & 1 & 1 \end{bmatrix}$. Since $|P| = -1 \neq 0$, then P^{-1} exists.

Therefore P diagonalizes the matrix A and $D = P^{-1}AP = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -3 \end{bmatrix}$.

4.3 Cayley-Hamilton Theorem

Theorem 4.9: Cayley-Hamilton Theorem

Every matrix is a zero of its characteristic polynomial. That is, $A \in M_n(\mathcal{R})$. Then,

$$p(\lambda) = \det(A - \lambda I_n) = 0 \Rightarrow p(A) = 0.$$

Example 4.9: Verify Cayley-Hamilton Theorem for $A = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$.

Solution:

$$p(\lambda) = \det(A - \lambda I_n) = \begin{vmatrix} 3-\lambda & 4 \\ 1 & 2-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (3-\lambda)(2-\lambda) - 4 = 0$$

$$\Rightarrow \lambda^2 - 5\lambda + 2 = 0$$

$$A^2 - 5A + 2I_2 = \begin{bmatrix} 13 & 20 \\ 5 & 8 \end{bmatrix} - 5 \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} + 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0. \text{ Therefore, } p(A) = 0.$$

Application of Cayley-Hamilton Theorem

1. Cayley-Hamilton Theorem may use to determine $A^r, r \geq 2$.

Example 4.10:

Form Example 3.27, $A^2 = 5A - 2I_2$.

$$\therefore A^3 = A^2(A) = (5A - 2I_2)A = 5A^2 - 2A = 5(5A - 2I_2) - 2A = 23A - 10I_2.$$

$$\therefore A^4 = A^3(A) = (23A - 10I_2)A = 23(5A - 2I_2) - 10A = 115A - 56I_2.$$

In general,
$$A^r = a_0 I_n + a_1 A, \quad r \geq 2. \quad \text{-----} (*)$$

Each of the eigenvalues λ of A is also satisfies the equation (*).

That is,
$$\lambda^r = a_0 + a_1 \lambda, \quad r \geq 2.$$

Determination of a_0 and a_1 :

Let $A \in M_2$. $p(\lambda) = \lambda^2 + c_1 \lambda + c_2 = 0$.

Case 1: $\lambda_1 \neq \lambda_2 \Rightarrow \lambda_1^r = a_0 + a_1 \lambda_1, \quad (r \geq 2) \quad \text{-----} (1)$

and $\lambda_2^r = a_0 + a_1 \lambda_2. \quad \text{-----} (2)$

(Solve the equations (1) and (2) for a_0 and a_1).

Case 2: $\lambda = \lambda_1 = \lambda_2 \Rightarrow \lambda^r = a_0 + a_1 \lambda \quad \text{-----} (3)$

and $r\lambda^{r-1} = a_1. (r \geq 2) \quad \text{-----} (4)$

(Solve the equations (3) and (4) for a_0 and a_1).

The technique in Case 2 is followed by the Theorem 3.5.

Theorem 4.10:

If $\lambda = \lambda_0$ are two (or more) repeated roots of the polynomial equation $f(\lambda) = 0$, then λ_0 is also a root $f'(\lambda) = 0$.

Proof:

Since $\lambda = \lambda_0$ are two repeated roots of $f(\lambda) = 0$, then $f(\lambda) = (\lambda - \lambda_0)(\lambda - \lambda_0)g(\lambda) = (\lambda - \lambda_0)^2 g(\lambda)$

Differentiate f with respect to λ , we have $f'(\lambda) = 2(\lambda - \lambda_0)g(\lambda) + (\lambda - \lambda_0)^2 g'(\lambda)$

$$\Rightarrow f'(\lambda_0) = 0 \Rightarrow \lambda_0 \text{ is a root } f'(\lambda) = 0.$$

Example 4.11:

If $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$. Determine A^5 and $A^r, r \geq 2$. Hence, evaluate $A^4 - 2A^3 + A - I$.

Solution:

$$|A - \lambda I_2| = \begin{vmatrix} 0 - \lambda & 1 \\ -2 & -3 - \lambda \end{vmatrix} = 0 \Rightarrow (-\lambda)(-3 - \lambda) + 2 = 0$$

$$\Rightarrow \lambda^2 + 3\lambda + 2 = 0 \Rightarrow (\lambda + 2)(\lambda + 1) = 0 \Rightarrow \lambda = -1, -2.$$

Let $A^5 = a_0 I_2 + a_1 A.$

For each $\lambda = -1, -2$, $\lambda^5 = a_0 + a_1 \lambda.$

When $\lambda = -1$, $(-1)^5 = a_0 + a_1(-1). \quad \text{-----} (1)$

When $\lambda = -2$, $(-2)^5 = a_0 + a_1(-2). \quad \text{-----} (2)$

Solve (1) and (2), we have $a_1 = 31$ and $a_0 = 30$.

$$\therefore A^5 = 30 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + 31 \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} 30 & 31 \\ -62 & -63 \end{bmatrix}.$$

$$\text{In general, } \lambda_1^r = a_0 + a_1 \lambda_1 \Rightarrow (-1)^r = a_0 + a_1(-1). \quad \text{-----(3)}$$

$$\lambda_2^r = a_0 + a_1 \lambda_2 \Rightarrow (-2)^r = a_0 + a_1(-2). \quad \text{-----(4)}$$

$$(3) - (4), \quad a_1 = (-1)^r - (-2)^r = (-1)^r (1 - 2^r). \quad \text{-----(5)}$$

$$\text{From (4) and (5), } a_0 = (-1)^r (1 - 2^r) + (-1)^r = (-1)^r (2 - 2^r).$$

$$\therefore A^r = ((-1)^r (2 - 2^r)) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + ((-1)^r (1 - 2^r)) \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} = (-1)^r \begin{bmatrix} 2 - 2^r & 1 - 2^r \\ 2^{r+1} - 2 & 2^{r+1} - 1 \end{bmatrix}.$$

$$\text{When } r = 4, \quad A^4 = (-1)^4 \begin{bmatrix} 2 - 2^4 & 1 - 2^4 \\ 2^{4+1} - 2 & 2^{4+1} - 1 \end{bmatrix} = \begin{bmatrix} 14 & -15 \\ 30 & 31 \end{bmatrix}.$$

$$\text{When } r = 3, \quad A^3 = (-1)^3 \begin{bmatrix} 2 - 2^3 & 1 - 2^3 \\ 2^{3+1} - 2 & 2^{3+1} - 1 \end{bmatrix} = \begin{bmatrix} 6 & 7 \\ -14 & -15 \end{bmatrix}.$$

$$\therefore A^4 - 2A^3 + A - I = \begin{bmatrix} 14 & -15 \\ 30 & 31 \end{bmatrix} - 2 \begin{bmatrix} 6 & 7 \\ -14 & -15 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -27 & -28 \\ 56 & 57 \end{bmatrix}.$$

Alternative method:

$$\text{Let } A^4 - 2A^3 + A - I = \beta_0 I + \beta_1 A.$$

$$\text{For each } \lambda = -1, -2, \quad \lambda^4 - 2\lambda^3 + \lambda - 1 = \beta_0 + \beta_1 \lambda.$$

$$\text{When } \lambda = -1, \quad (-1)^4 - 2(-1)^3 + (-1) - 1 = \beta_0 + \beta_1(-1). \quad \text{-----(6)}$$

$$\text{When } \lambda = -2, \quad (-2)^4 - 2(-2)^3 + (-2) - 1 = \beta_0 + \beta_1(-2). \quad \text{----- (7)}$$

Solve (6) and (7), we have $\beta_0 = -27$ and $\beta_1 = -28$.

$$\therefore A^4 - 2A^3 + A - I = -27 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 28 \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} = \begin{bmatrix} -27 & -28 \\ 56 & 57 \end{bmatrix}.$$

Example 4.12:

If $A = \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix}$. Determine A^5 and $A^r, r \geq 2$. Hence, evaluate $A^4 - 2A^3 + A - I$.

Solution:

$$\det(A - \lambda I_2) = \begin{vmatrix} 0 - \lambda & 1 \\ -1 & -2 - \lambda \end{vmatrix} = 0 \Rightarrow (-\lambda)(-2 - \lambda) + 1 = 0 \Rightarrow \lambda^2 + 2\lambda + 1 = 0$$

$$\Rightarrow (\lambda + 1)^2 = 0 \Rightarrow \lambda = -1 \text{ (repeated)}.$$

$$\text{Let } A^5 = a_0 I_2 + a_1 A.$$

$$\text{For } \lambda = -1, \quad \lambda^5 = a_0 + a_1 \lambda \quad \text{and} \\ \text{the derivative is } 5\lambda^4 = a_1.$$

$$\begin{aligned} \text{When } \lambda = -1, \quad (-1)^5 &= a_0 + a_1(-1). & \text{----- (1)} \\ \text{and } 5(-1)^4 &= a_1. & \text{----- (2)} \end{aligned}$$

Solve (1) and (2), we have $a_1 = 5$ and $a_0 = 4$.

$$\therefore A^5 = 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + 5 \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} 4 & 5 \\ -5 & -6 \end{bmatrix}.$$

$$\begin{aligned} \text{Let } A^4 - 2A^3 + A - I &= \beta_0 I + \beta_1 A. \\ \text{For each } \lambda = -1, \quad \lambda^4 - 2\lambda^3 + \lambda - 1 &= \beta_0 + \beta_1 \lambda. \\ \text{the derivative is } 4\lambda^3 - 6\lambda^2 + 1 &= \beta_1. \end{aligned}$$

$$\begin{aligned} \text{When } \lambda = -1, \quad (-1)^4 - 2(-1)^3 + (-1) - 1 &= \beta_0 + \beta_1(-1). & \text{----- (3)} \\ \text{and } 4(-1)^3 - 6(-1)^2 + 1 &= \beta_1. & \text{----- (4)} \end{aligned}$$

Solve (3) and (4), we have $\beta_1 = -9$ and $\beta_0 = -8$.

$$\therefore A^4 - 2A^3 + A - I = -8 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - 9 \begin{bmatrix} 0 & 1 \\ -1 & -2 \end{bmatrix} = \begin{bmatrix} -8 & -9 \\ 9 & 10 \end{bmatrix}.$$

2. Cayley-Hamilton Theorem may use to determine A^{-1} , the inverse of A .

Example 4.13:

$$\text{Let } A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}. \text{ Use the Cayley Hamilton Theorem to find the inverse of } A.$$

Solution:

From Example 4.3, the characteristic equation is given by

$$p(\lambda) = \det(A - \lambda I_n) = \lambda^3 - 2\lambda^2 - \lambda + 2 = 0.$$

By the Cayley-Hamilton Theorem, the matrix A satisfies

$$p(A) = A^3 - 2A^2 - A + 2I_3 = 0.$$

$$\begin{aligned} & A^3 - 2A^2 - A + 2I_3 = 0 \\ \Rightarrow & A^{-1}(A^3 - 2A^2 - A + 2I_3) = A^{-1}(0) \\ \Rightarrow & (A^{-1}A)A^2 - 2(A^{-1}A)A - A^{-1}A + 2A^{-1} = 0 \\ \Rightarrow & A^2 - 2A - I_3 + 2A^{-1} = 0 \\ \Rightarrow & A^{-1} = \frac{1}{2}(-A^2 + 2A + I_3) \\ \Rightarrow & A^{-1} = \frac{1}{2} \left\{ - \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix} + 2 \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\} \\ & = \frac{1}{2} \left\{ - \begin{bmatrix} 0 & 1 & 1 \\ -3 & 4 & 3 \\ -1 & 1 & 2 \end{bmatrix} + 2 \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\} = \begin{bmatrix} 3 & 1 & -5 \\ 1 & 1 & -1 \\ 1 & 1 & -3 \end{bmatrix}. \end{aligned}$$